Letter Of Transmittal

Degree Project Supervisor
Bachelor of Chemical Engineering
Lakehead University

Date: November 9, 2026

Dear Dr. Pakzad and Dr. Rakshit,

Please find attached our first preliminary report for our degree project, *Hydrometallurgical Process Design for Critical Material Recovery from Black Mass Feedstock*. The report outlines our initial findings on processing NMC 111 black mass to recover dried Lithium Carbonate, along with our process objectives and next steps.

We welcome any feedback or suggestions to help improve our final report and appreciate your continued support and guidance throughout this project.

Sincerely,

Sonali Brighton
sbright1@lakeheadu.ca

Harshi Dave
hmdave@lakeheadu.ca

Stephen Jones
sdjones1@lakeheadu.ca

Nathan Ladouceur
ndladouc@lakeheadu.ca

Ammar Rajeh
arajeh1@lakeheadu.ca

Hydrometallurgical Process Design for Critical Material Recovery from Black Mass Feedstock

Authors: Sonali Brighton, Harshi Dave, Stephen Jones, Nathan Ladouceur, Ammar Rajeh

Supervisors: Leila Pakzad, Sudip Rakshit

Report Submitted: November 9th, 2026.



Department of Chemical Engineering

Lakehead University

955 Oliver Rd, Thunder Bay, ON P7B 5E1

Table of Contents

I. Summary 7

II. Introduction 8

Table 1: Global and Canadian Lithium-Ion Battery Market Values and Growth Projections (2025–2035). 11

Table 2. Comparison of Different Methods for Recovering Valuable Materials from Li-ion Battery Black Mass. 13

Table 3. Process Stream Identification and Descriptions for the Hydrometallurgical Recovery of Valuable Metals from NMC111 Black Mass. 16

Table 4. Service Stream Identification and Descriptions for Utilities in the Hydrometallurgical Recovery Process. 18

Table 5. Operating Conditions and Stream Information for the Chemical Leaching Reactors (R-101A/B/C/D). [3] 19

Table 6. Leaching Efficiencies for the Chemical Leaching Process. 23

Table 7. Operating Conditions and Stream Information for the Decanter Centrifugation Units (C-101A/B/C/D). 27

Table 8. Solid-Liquid Separation and Washing Efficiencies for the Decanter Centrifugation Unit. [10] [11] 28

Table 9. Operating Conditions and Stream Information for the Copper Sulfide Precipitation Units (R-102A/B). [14] 35

Table 10. Copper Precipitation and Recovery Efficiencies. [14] 36

Table 11. Operating Conditions and Stream Information for the Nitrogen Sparging Unit (V-101A/B). [16] 39

Table 12. Hydrogen Sulfide Removal Efficiencies for the Nitrogen Sparging Process. 39

Table 13. Operating Conditions and Stream Information for the Iron and Aluminum Precipitation Unit (R-103A/B). 44

Table 14. Iron and Aluminum Precipitation Efficiencies. 45

Table 15. Operating Conditions and Stream Information for the NMC Oxalate Precipitation Unit (R-104A/B). [24] [26] 48

Table 16. NMC Oxalate Precipitation and Separation Efficiencies. [3] [25] [34] 49

Table 17. Operating Conditions and Stream Information for the NMC Vacuum Filtration Unit (VF-101A/B) 53

Table 18. Combined NMC Oxalate Precipitation and Filtration Recoveries. [34] 54

Table 19. Operating Conditions and Stream Information for the Lithium Carbonate Precipitation Unit (R-105A/B). [4] [27] [28] 58

III. Project Statement 61

IV. Thermodynamics & Kinetics 62

V. Process Flow Diagram 66

VI. Mass & Energy Balances 68

Mass Balance 68

Table 20. Summary of Process Stream Descriptions, Compositions, and Total Mass Flow Rates 68

Energy Balance 72

Table 21. Energy Balance Calculation Summary 75

VII. Challenges 76

VIII. Conclusions & Future Works 80

Acknowledgement 80

Nomenclature 81

***Table 22*.** List of Abbreviations, Notations & Chemical Symbols, and Greek Letters & Mathematical Symbols and Their Definitions Used in the Hydrometallurgical Process Design Report. 81

IX. References and Citations 83

List of Figures and Tables

***Figure 1*:** Electric vehicle and stationary storage battery deployment by battery production location, 2018-2025 [1]. 9
Figure 1. Process Flow Diagram for The Hydrometallurgical Recovery of Critical Materials from Spent Lithium-ion Battery Black Mass. 14
Figure 2. Process Flow Diagram of the Chemical Leaching Reactor Unit (R-101A/B/C/D). 19
Figure 3: Leaching Yield of Metals from Different Cathode Materials at Operating Conditions. 21
Figure 4: Leaching of Metals from a Black Mass Sample ($Li_{1.09}Ni_{0.31}Mn_{0.39}Co_{0.39}O_2$) at Operating Conditions. 21
Figure 5: Leaching yield of metals from different cathode materials using two methods INIT and MULT. 22
Figure 6. Process Flow Diagram of the Decanter Centrifugation Unit (C-101A/B/C/D). 26
***Figure 7*.** Process Flow Diagram of the Copper Sulfide Precipitation Unit (R-102A/B). 34
Figure 8. Process Flow Diagram of the Nitrogen Sparging Unit (V-101A/B). 38
Figure 9. Process Flow Diagram of the Iron and Aluminium Precipitation Unit (R-103A/B). 43
Figure 10. Process Flow Diagram of the NMC Oxalate Precipitation Unit (R-104A/B). 47
NMC oxalate precipitation is a chemical separation technique that uses oxalic acid to selectively recover nickel, cobalt, and manganese from the metal-rich leach solution. The unit is described as an agitated precipitation reactor. Oxalic acid reacts with the dissolved metal ions to form insoluble metal oxalates, which precipitate as suspended solids. The resulting mixture is then subjected to solid-liquid separation through vacuum filtration, producing an NMC oxalate-rich filter cake and a lithium-rich filtrate. The performance process depends on operating conditions such as temperature, pH, reagent concentration, reaction time, and mixing intensity. 47
Figure 11. Process Flow Diagram of the NMC Vacuum Filtration Unit (VF-101A/B). 52
Figure 12. Process Flow Diagram of the Lithium Carbonate Precipitation Unit (R-105A/B). 57
Figure 13. Process Flow Diagram of the Hydrometallurgical Process for NMC 111 Black Mass (Reproduced from Figure 1) 67

Table 1: Global and Canadian Lithium-Ion Battery Market Values and Growth Projections (2025–2035). 11
Table 2. Comparison of Different Methods for Recovering Valuable Materials from Li-ion Battery Black Mass. 13
***Table 3*.** Process Stream Identification and Descriptions for the Hydrometallurgical Recovery of Valuable Metals from NMC111 Black Mass. 16
***Table 4*.** Service Stream Identification and Descriptions for Utilities in the Hydrometallurgical Recovery Process. 18
Table 5. Operating Conditions and Stream Information for the Chemical Leaching Reactors (R-101A/B/C/D). [3] 19
Table 6. Leaching Efficiencies for the Chemical Leaching Process. 23
Table 7. Operating Conditions and Stream Information for the Decanter Centrifugation Units (C-101A/B/C/D). 27
Table 8. Solid-Liquid Separation and Washing Efficiencies for the Decanter Centrifugation Unit. [10] [11] 28
Table 9. Operating Conditions and Stream Information for the Copper Sulfide Precipitation Units (R-102A/B). [14] 35
Table 10. Copper Precipitation and Recovery Efficiencies. [14] 36
Table 11. Operating Conditions and Stream Information for the Nitrogen Sparging Unit (V-101A/B). [16] 39
Table 12. Hydrogen Sulfide Removal Efficiencies for the Nitrogen Sparging Process. 39
Table 13. Operating Conditions and Stream Information for the Iron and Aluminum Precipitation Unit (R-103A/B). 44
Table 14. Iron and Aluminum Precipitation Efficiencies. 45
Table 15. Operating Conditions and Stream Information for the NMC Oxalate Precipitation Unit (R-104A/B). [24] [26] 48
Table 16. NMC Oxalate Precipitation and Separation Efficiencies. [3] [25] [34] 49
Table 17. Operating Conditions and Stream Information for the NMC Vacuum Filtration Unit (VF-101A/B) 53
Table 18. Combined NMC Oxalate Precipitation and Filtration Recoveries. [34] 54
Table 19. Operating Conditions and Stream Information for the Lithium Carbonate Precipitation Unit (R-105A/B). [4] [27] [28] 58
Table 20. Summary of Process Stream Descriptions, Compositions, and Total Mass Flow Rates 68
Table 21. Energy Balance Calculation Summary 75
***Table 22*.** List of Abbreviations, Notations & Chemical Symbols, and Greek Letters & Mathematical Symbols and Their Definitions Used in the Hydrometallurgical Process Design Report. 81

I. Summary

This degree project investigates the development of a preliminary hydrometallurgical process for recovering critical materials from NMC (nickel-manganese-cobalt) black mass feedstock. The current research focuses on NMC (nickel-manganese-cobalt) feedstock and develops a preliminary process concept based on mechanical/pre-treatment requirements followed by chemical leaching, solid-liquid separation, purification, precipitation, and drying.

Hydrometallurgy when compared to pyrometallurgy was selected as the principal recovery route because it offers high metal recover, selective recover capability, low energy consumption, and reduced environmental impact relative to conventional recovery routes.

The preliminary recovery route removes aluminum by NaOH precipitation, recovers nickel, cobalt, and manganese through solvent extraction and precipitation, and recovers lithium from the remaining solution as lithium carbonate. Iron and copper are treated as impurity streams requiring removal before final battery-material recovery.

II. Introduction

Lithium-ion batteries are becoming more important every day for vehicles and other low-carbon technologies. When these batteries reach the end of their life recycling them helps recover key materials that would otherwise need to be mined through energy-intensive processes. In this literature review spent batteries are defined as those that're no longer useful and the main materials needed for low-carbon technologies are clearly identified.

Research into lithium-ion batteries really started after the oil crisis in 1973–1974. Countries like Europe and Japan which depended heavily on oil imports wanted to reduce their risks. They began investing in battery technologies that could power vehicles and store energy from renewable sources. The big breakthroughs came over the few decades. In the 1970s M. Stanley Whittingham at Exxon created the rechargeable lithium battery. It used a titanium disulfide cathode and a metallic lithium anode. Then in 1980 John B. Goodenough at the University of Oxford found that lithium cobalt oxide ($LiCoO_2$) could be used as a cathode material. This doubled the energy capacity of the battery. In 1985 Akira Yoshino and Asahi Kasei made another step by replacing the unstable metallic lithium anode with petroleum coke. This change made the battery much safer. Laid the foundation for today's lithium-ion cells.

Japanese electronics companies were among the first to see the potential. They got a patent for a prototype in 1986. By 1991 Sony and Asahi Kasei brought the lithium-ion battery to market. It was used in consumer electronics.

Soon the technology moved into cars. In 1996 Nissan built the Prairie Joy, which was the electric vehicle to use lithium-ion batteries. In 2008 Tesla, an automaker released the Roadster—a high-

performance electric car that showed what lithium-ion batteries could do. Then in 2010 Nissan launched the Leaf. It became the electric vehicle to sell more than 100,000 units by 2013.

The growth in lithium-ion manufacturing has been huge. In 2019 the Nobel Prize in Chemistry was awarded to Whittingham, Goodenough and Yoshino for their work. Their contributions led to an increase in the number of batteries being made. As a result, there is now a growing number of end-of-life batteries. To secure the supply of materials for future low-carbon technologies, advanced recycling methods, like hydrometallurgy must be adopted. It is no longer optional—it is necessary.

Figure I shows an EV vehicle and stationary storage comparison between among Chinese, European, and North American markets.

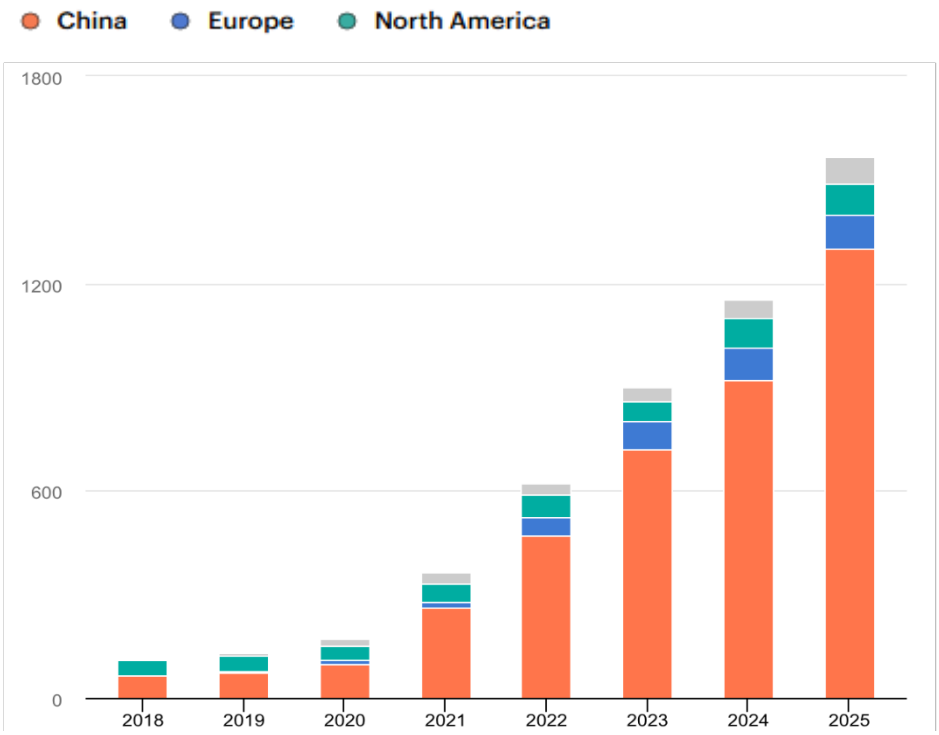


***Figure I*:** Electric vehicle and stationary storage battery deployment by battery production location, 2018-2025 [1].

Lithium-ion batteries are a critical component in pushing toward a more sustainable future. They power industrial sensor modules and robotics to advance manufacturing as part of Industry 4.0 implementations. Beyond everyday portable electronics like phones, tablets, and laptops, these batteries are essential for large-scale energy storage. By integrating power generated from wind and solar, they ensure a steady electricity supply for homes, hospitals, and municipal grids. Ultimately, lithium-ion batteries provide an environmentally friendly power supply for modern transportation, making it easier to plug clean energy directly into daily life. [2].

EV batteries are undergoing rapid expansion, and the global lithium-ion battery market is valued around $134.8 billion to $151.0 billion in 2025. It is projected that it will reach between $166.2 billion and $174.9 billion in 2026. New chemistries are emerging and resulting in new formations to formulate. Lithium Cobalt Oxide and Lithium Iron Phosphate account for the largest shares in all lithium-ion applications. The Asia-Pacific region controls majority of global battery production. North America and Europe however are recognized as the fastest growing regions. High suitability for electric transportation and the need to combat climate changes ensure sustained global demand for lithium-ion energy storage. Estimates by the McKinsey Battery Insights team indicate that the complete lithium-ion value chain could achieve a compound annual growth rate exceeding 30% from 2022 to 2030, attaining a 4.7 TWh capacity and a market valuation above $400 billion.

Canada is rapidly progressing through the EV route. Canada's domestic lithium-ion battery market is valued approximately $1.19 billion in 2025 and is expected to grow at a CAGR of 10.8% through 2030. This rate currently outpaces the global average baseline for localized national markets. ***Table 1*** presents a comparison of the global and Canadian lithium-ion battery markets, highlighting their estimated market values in 2025 and projected growth through 2035.

Table 1: Global and Canadian Lithium-Ion Battery Market Values and Growth Projections (2025–2035).

	2025 Value (USD)	2026 Forecast (USD)	2030-2035 Projections (USD)
Global Li-ion Market	$134.8B-151.0B	$166.2B-$174.9B	$405.4B (2033) $887.8B (2035)
Canada Li-ion Market	$1.19B	$1.3B	$2.69B (2030)

The recovery process consists of several key separation and recovery steps. Solid-liquid extraction (chemical leaching) is used to transfer valuable metals from the solid black mass into a liquid solution using sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2). The resulting mixture is then subjected to solid-liquid separation, such as centrifugation, to remove the remaining solid material, primarily graphite. Selective precipitation is subsequently used to purify and recover the dissolved metals through the removal of impurities, including copper, iron, and aluminum, followed by the precipitation of nickel, cobalt, and manganese as metal oxalates.

The remaining lithium-rich solution is then processed through precipitation and crystallization to recover lithium as lithium carbonate (Li_2CO_3). The recovered solid products undergo washing and drying to remove residual impurities and moisture and obtain the final products.

Table 2 presents different methods for the recovery of valuable materials from spent lithium-ion battery black mass. Pyrometallurgy can provide rapid metal recovery and can process different battery chemistries, but it requires high operating temperatures and may generate harmful gaseous emissions. Direct recycling aims to restore cathode materials directly, reducing the need to convert materials into individual metal products; however, the process can be sensitive to variations in battery chemistry and material composition. Bioleaching operates under relatively mild conditions

and can reduce chemical consumption and energy requirements, but its slow kinetics and scale-up challenges can limit its application for large-scale processing.

Hydrometallurgy uses aqueous solutions to leach and separate valuable metals and offers the potential for selective recovery at relatively low temperatures. However, it requires chemical reagents, pre-treatment, and appropriate wastewater management. Based on these characteristics, hydrometallurgy was selected as the primary recovery approach for this project because it provides a suitable route for selectively separating and recovering nickel, cobalt, manganese, and lithium from NMC black mass.

Table 2. Comparison of Different Methods for Recovering Valuable Materials from Li-ion Battery Black Mass.

Method	Description	Main Advantages Identified	Main Disadvantages Identified
Hydrometallurgy	1. Acid leaching transfers valuable metals from cathode/black mass material into solution for separation.	1. Selective recovery. 2. Low energy consumption. 3. Good extraction capability.	1. Can be resource-intensive and requires pre-treatment. 2. Wastewater management may be required.
Pyrometallurgy	2. Smelting at 500-2000 °C is used to recover metals.	1. Fast recovery 2. Limited pre-treatment. 3. Can handle different LIB chemistries.	1. Resource intensive. 2. Can produce toxic gases.
Direct Recycling	1. Cathode-to-cathode restoration rather than recovery of the constituent materials.	1. Potentially economically feasible recovery of battery components.	1. Recovered material will not behave like virgin material. 2. Mixing of cathode can lower product quality. 3. Primarily a lab scale process and not yet widely industry applied.
Bioleaching	1. Microorganisms produce acids that dissolve valuable metals.	1. Lower chemical use. 2. Low energy consumption. 3. Mild operating temperatures. 4. Fewer harmful emissions.	1. Slow kinetics. 2. Specific living conditions. 3. Lower processing rate 4. Difficult to scale up.

The proposed hydrometallurgical recovery process consists of several interconnected unit operations designed to selectively extract, separate, and recover valuable metals from spent NMC111 lithium-ion battery black mass. The process begins with chemical leaching using sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2) to dissolve the target metals, followed by solid-liquid separation to remove undissolved residues, primarily graphite. The resulting metal-rich leach

solution undergoes successive purification and precipitation stages to remove impurities and selectively recover nickel, cobalt, and manganese as mixed metal oxalates, followed by lithium recovery as lithium carbonate (Li_2CO_3). Subsequent filtration, washing, and drying operations are employed to isolate and prepare the recovered products. ***Figure 1*** presents the process flow diagram illustrating the sequence of major unit operations involved in the proposed hydrometallurgical recovery process.

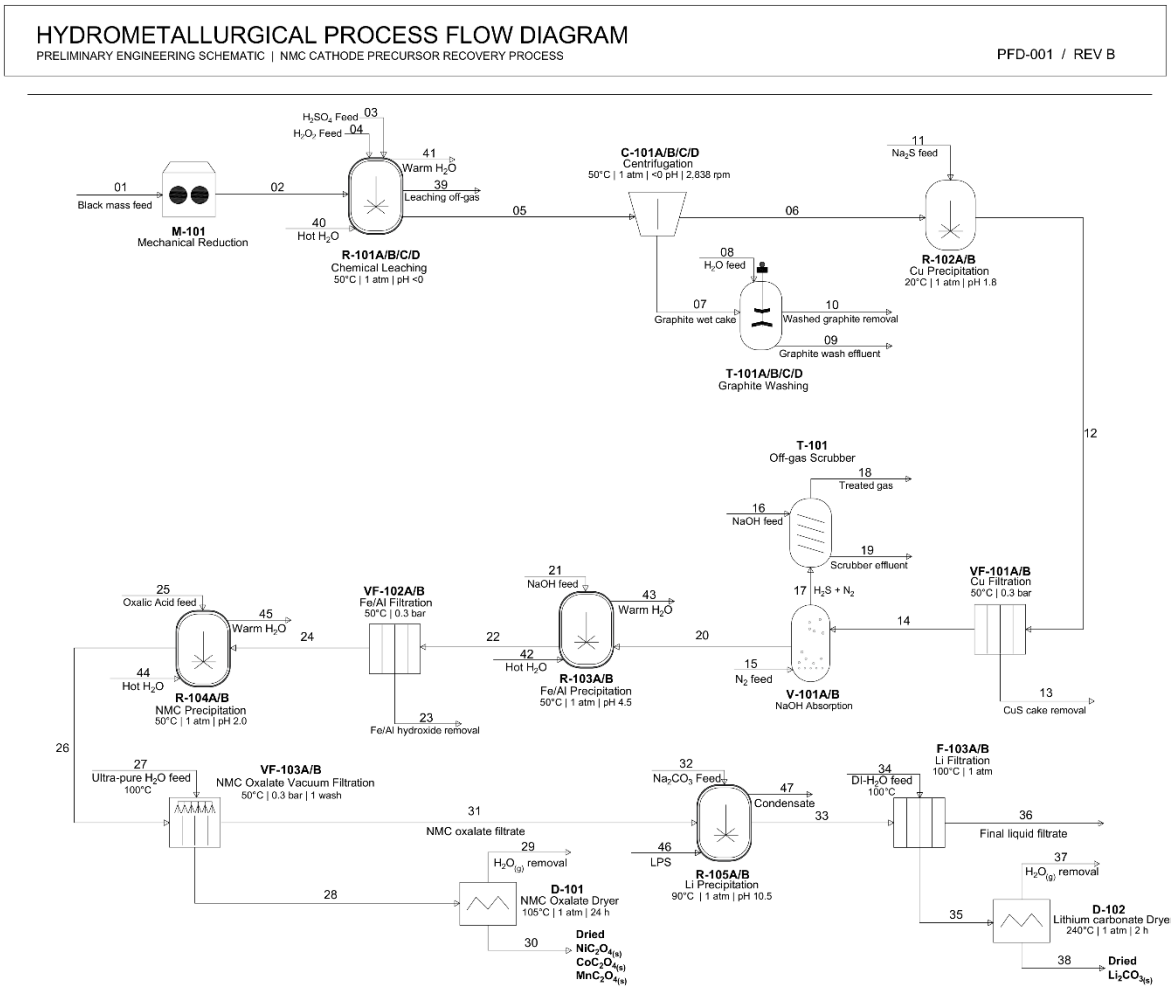


Figure 1. Process Flow Diagram for The Hydrometallurgical Recovery of Critical Materials from Spent Lithium-ion Battery Black Mass.

Additionally, ***Table 3*** provides a comprehensive list of the all-process streams associated with the proposed hydrometallurgical recovery of valuable metals from NMC111 lithium-ion battery black mass, as illustrated in ***Figure 1***. These streams include raw materials, chemical reagents, intermediate process streams, recovered products, and waste streams. The streams are organized according to the major processing stages, including black mass preparation, chemical leaching, graphite separation, copper removal, off-gas treatment, iron and aluminum removal, NMC oxalate recovery, and lithium carbonate recovery. This stream identification system establishes a consistent reference for conducting material and energy balances, tracking material flows, and evaluating individual unit operations throughout the process.

Table 3. Process Stream Identification and Descriptions for the Hydrometallurgical Recovery of Valuable Metals from NMC111 Black Mass.

Stream Number	Process Section (Key)	Stream Description
1	Black Mass Feed Preparation	Black mass feed
2	Black Mass Feed Preparation	Mechanically reduced NMC111 black mass feed
3	Chemical Leaching	Sulfuric acid (H_2SO_4) feed
4	Chemical Leaching	Hydrogen peroxide (H_2O_2) feed
5	Chemical Leaching	Leaching slurry to decanter centrifugation
6	Graphite Separation / Copper Precipitation	Metal-rich leach solution to copper precipitation
7	Graphite Separation	Graphite-rich wet cake
8	Graphite Washing	Water feed for graphite washing
9	Graphite Washing	Graphite washes effluent
10	Graphite Washing	Washed graphite removal
11	Copper Precipitation	Sodium sulfide (Na_2S) feed
12	Copper Precipitation / Filtration	Copper sulfide (CuS) slurry
13	Copper Filtration	Copper sulfide (CuS) filter cake removal
14	Copper Filtration / Nitrogen Sparging	Copper-depleted filtrate to nitrogen sparging
15	Nitrogen Sparging	Nitrogen (N_2) feed
16	Off-Gas Scrubbing	Sodium hydroxide (NaOH) feed
17	Nitrogen Sparging / Off-gas Scrubbing	Hydrogen sulfide (H_2S) and nitrogen (N_2) Off-gas
18	Off-gas Scrubbing	Treated Off-gas
19	Off-gas Scrubbing	Scrubber effluent
20	Nitrogen Sparging / Fe-Al Precipitation	Sparged leach solution to iron/aluminum precipitation
21	Fe/Al Precipitation / Filtration	Sodium hydroxide (NaOH) feed
22	Fe/Al Precipitation / Filtration	Iron/aluminum hydroxide precipitation slurry

23	Fe/Al Filtration	Iron/aluminum hydroxide solids removal
24	Fe/Al Filtration / NMC Precipitation	Purified filtrate to NMC oxalate precipitation
25	NMC Precipitation	Oxalic acid feed
26	NMC Precipitation / Vacuum Filtration	Mixed nickel, cobalt, and manganese oxalate slurry
27	NMC Oxalate Vacuum Filtration	Ultrapure water feed for NMC oxalate cake washing
28	NMC Oxalate Vacuum Filtration / Drying	Wet mixed NMC oxalate filter cake
29	NMC Oxalate Drying	Water vapour removed during NMC oxalate drying
30	NMC Oxalate Drying	Dried nickel, cobalt, and manganese oxalate product
31	NMC Oxalate Vacuum Filtration / Li Precipitation	Lithium-rich NMC oxalate filtrate
32	Lithium Precipitation	Sodium carbonate (Na_2CO_3) feed
33	Lithium Precipitation / Filtration	Lithium carbonate precipitation slurry
34	Lithium Filtration	Deionized water feed for lithium carbonate washing
35	Lithium Filtration / Drying	Wet lithium carbonate filter cake
36	Lithium Filtration	Final liquid filtrate
37	Lithium Carbonate Drying	Water vapour removed during lithium carbonate drying
38	Lithium Carbonate Drying	Dried lithium carbonate (Li_2CO_3) product
39	Chemical Leaching	Leaching Off-gas

Table 4. Service Stream Identification and Descriptions for Utilities in the Hydrometallurgical Recovery Process.

Stream Number	Process Section (Key)	Stream Description
40	Chemical Leaching (R-101A/B/C/D)	Hot H_2O feed to reactor heating jacket
41	Chemical Leaching (R-101A/B/C/D)	Warm H_2O leaving reactor heating jacket
42	Fe/Al Precipitation (R-103A/B)	Hot H_2O feed to reactor heating jacket
43	Fe/Al Precipitation (R-103A/B)	Warm H_2O leaving reactor heating jacket
44	NMC Precipitation (R-104A/B)	Hot H_2O feed to reactor heating jacket
45	NMC Precipitation (R-104A/B)	Warm H_2O leaving reactor heating jacket
46	Li Precipitation (R-105A/B)	LPS feed to reactor heating jacket
47	Li Precipitation (R-105A/B)	Condensed H_2O leaving reactor heating jacket

1. Chemical Leaching Reactor Units

2. Purpose

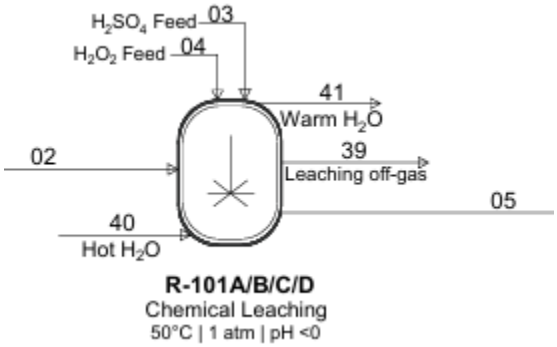


Figure 2. Process Flow Diagram of the Chemical Leaching Reactor Unit (R-101A/B/C/D).

Chemical Leaching is the process in which a liquid solvent extracts the soluble components out of a solid material. By using a chemical leaching reactor, it will carry out the extraction process by the addition of sulfuric acid (H_2SO_4). Chemical leaching involves a chemical reaction between the solid and the leaching agent which is known as the lixiviant to dissolve the target substance. For our process the chemical leaching stage will dissolve the critical metals, Lithium, Nickel, and Cobalt, from the black mass into an aqueous pregnant leach solution (PLS). [3] [4] [5]

a. Conditions

Table 5. Operating Conditions and Stream Information for the Chemical Leaching Reactors (R-101A/B/C/D). [3]

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	NMC111 black mass, sulfuric acid (H_2SO_4), and hydrogen peroxide (H_2O_2).

Product(s) out	Pregnant leach solution (PLS), containing dissolved metals and suspended undissolved solids.
Temperature (°C)	50
Pressure (atm)	1.0
pH	< 0
Leaching Acid	2.0 M H_2SO_4
Reducing Agent	3.0% v/v H_2O_2
Solid-to-Liquid (S/L) Ratio (g/mL)	1:20
Retention Time (min)	60
Agitation Rate (rpm)	300

b. Reactions/Kinetics

Primary Chemical Reactions:

1. $LiNi_xMn_yCo_zO_{2(s)} + \frac{3}{2}H_2SO_{4aq} + \frac{1}{6}H_2O_{2(aq)} \rightarrow \frac{1}{3}NiSO_{4(aq)} + \frac{1}{3}MnSO_{4(aq)} + \frac{1}{3}CoSO_{4(aq)} + \frac{1}{2}Li_2SO_{4(aq)}$ **[Equation 1]**

2. $2LiCoO_{2(s)} + 3H_2SO_{4(aq)} + H_2O_{2(aq)} \rightarrow 2CoSO_{4(aq)} + Li_2SO_{4(aq)} + 4H_2O_{(l)} + O_{2(g)}$ **[Equation 2]**

Side Reactions:

3. $3Cu^0_{(s)} + 2Co^{3+}_{(lattice)} \rightarrow 3Cu^{3+}_{(aq)} + 2Co^{2+}_{(aq)}$ **[Equation 3]**

4. $2Al^0_{(s)} + 6H^+_{(aq)} \rightarrow 2Al^{3+}_{(aq)} + 3H_{2(g)}$ **[Equation 4]** [3]

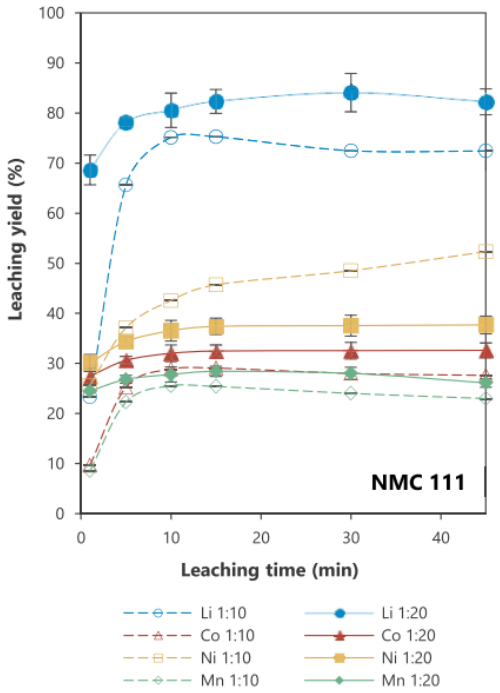


Figure 3: Leaching Yield of Metals from Different Cathode Materials at Operating Conditions.

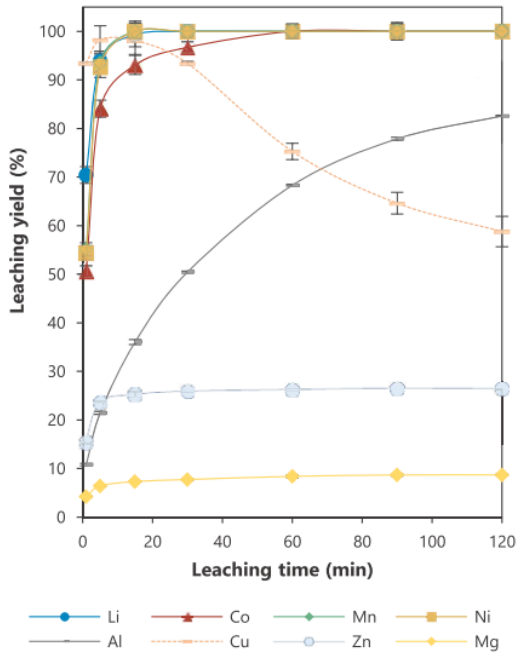


Figure 4: Leaching of Metals from a Black Mass Sample ($Li_{1.09}Ni_{0.31}Mn_{0.39}Co_{0.39}O_2$) at Operating Conditions.

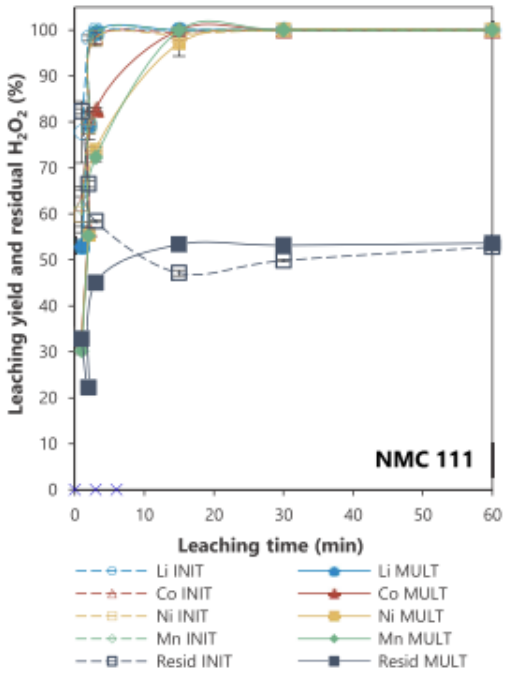


Figure 5: Leaching yield of metals from different cathode materials using two methods INIT and MULT.

The different methods are INIT which is adding the total volume of H_2O_2 at the beginning when $t = 0$. MULT is stagged additions of H_2O_2 [3.2]. Based on Figure 5 we can say that the reaction rate is higher when the total amount of hydrogen peroxide is added initially.

The leaching behavior of a real black mass sample from spent EV LIBs was evaluated. The fraction of Nickel, Manganese, and Cobalt is close to NMC 111 chemistry.

- From this figure we can see that Li, Ni, and Mn reach 100% dissolution within 15 minutes.
- Cobalt reaches 93% yield at 15 min and 100% dissolution at 60 minutes. Copper peaks rapidly 98% at 5 mins then decreases to about 60% after 120 minutes.

- Aluminium dissolves progressively reaching 35% at 15 minutes, 68% at 60 minutes, and 83% at 120 minutes. Zinc and Magnesium reach 25% and 7% at 15 minutes which level off at 26% and 9% at 120 minutes.

The overall mechanism which is occurring is the delithiation, the removal of lithium ions from an electrode. The release of the lithium ion which is driven by the structural bonding results in the fragile ionic bonds to connect lithium and oxygen in layered cathode structures and the stronger bonds are connected by transition metals and oxygen.

c. Separation Efficiencies

Table 6. Leaching Efficiencies for the Chemical Leaching Process.

Element/Component	**Efficiency (%)**
Lithium (Li)	100% within 15 minutes
Nickel (Ni)	100% within 15 minutes
Manganese (Mn)	100% within 15 minutes
Cobalt (Co)	93% at 15 minutes at 100% after 60 minutes
Copper (Cu)	98% within 5 minutes then drops to 59% at 120 minutes
Aluminium	36% within 15 minutes, 68% at 60 minutes, and 83% at 120 minutes
Zinc (Zn)	25% within 15 minutes and levels off at 26%.
Magnesium (Mg)	Reaches 7% separation yield at 15 minutes and levels off at 9%.
Graphite (Carbon, C)	100% insoluble

d. Advantages & Disadvantages

- **Advantages**:

1. High extraction efficiency as 100% of Lithium, Nickel, Cobalt, and Manganese is fully recovered.
2. Faster process kinetics as the metal dissolution occurs within 60 minutes.
3. Can be scaled industrially as standard chemical equipment and commodity reagents, sulfuric acid and hydrogen peroxide, is used.
4. Effective phase separation leaves the graphite intact for recovery when it goes through the filtration process.

- **Disadvantages**:

1. Up to 40-60% of added Hydrogen Peroxide remains unreacted.
2. Increase of impurity during chemical leaching as high extraction of the copper and aluminum.
3. Produces high concentration of aqueous sulfate ions requiring wastewater treatment.
4. Continuous oxygen generation creates risk of slurry foaming and over pressurization. [3] [4] [5]

e. Equipment Design

Continuous Stirred-Tank Reactors (CSTRs) will be operating in series to limit the fluid short-circuiting. As Lithium., Nickel, and Manganese dissolves within 15 minutes and Cobalt requires a full 60-minute retention time to achieve 100% dissolution if by any chance unreacted

solid $LiCoO_2$ or NMC oxide particles leave the reactor it will leave to a loss in metal recovery and economic yield. 1 CSTR will be there as a back-up. Further research needs to be done on the material and how the reactors will be specifically designed. [4] [5]

f. Waste Treatment & Process Concerns

Acidic Sulfate Wastewater will be generated in the chemical leaching process. A treatment process would require lime ($Ca(OH)_2$) neutralization to produce gypsum ($CaSO_4 * 2H_2O$) or mechanical vapor recompression (MVR) crystallization to produce industrial sodium sulfate (Na_2SO_4). [5] [6]

3. Decanter Centrifuge Units

a. Purpose

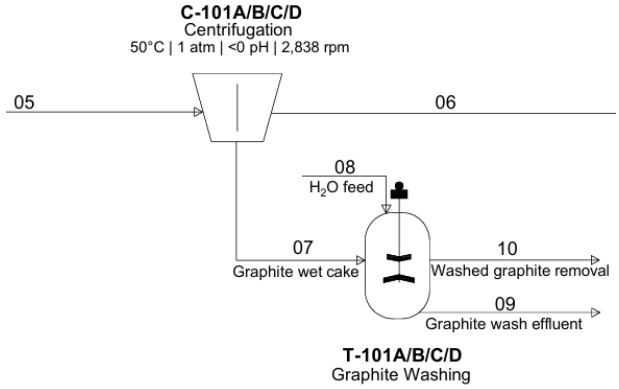


Figure 6. Process Flow Diagram of the Decanter Centrifugation Unit (C-101A/B/C/D).

Centrifugation is a solid–liquid separation technique that uses rapid rotation to generate a centrifugal force many times greater than gravity. Denser solid particles are driven towards the wall of the rotating bowl, while the lighter liquid remains closer to the axis of rotation. The mixture is thereby split into a clarified liquid (the centrate) and a concentrated solid (the cake). Because the process replaces slow gravitational settling with a strong, controllable force, separation that would take hours can be completed in seconds to minutes. Its performance depends on the density difference between the solid and the liquid, the particle size, the viscosity of the liquid, and the residence time in the machine. [7]

In this project, black mass from NMC111 lithium-ion batteries is leached to dissolve the valuable metals (Ni, Co, Mn, and Li). The slurry leaving the Chemical Leaching Reactor contains the metal-bearing pregnant leach solution (PLS) together with undissolved solid residue. It is fed to a decanter centrifuge, which separates the PLS from the residue based on density difference. [8] [9]

The residue consists mainly of graphite and Carbon (C) black, together with residual polyvinylidene fluoride (PVDF) binder ($C_2H_2F_2$), ZrO_2 (Zirconium Oxide), W (Tungsten)

particles, and possibly iron precipitates. It is discharged continuously as a cake, which is an advantage over batch filtration at an industrial scale. [8] [9]

b. Operating Conditions & Stream Information

The operating conditions and stream information for the decanter centrifugation process are summarized in ***Table 7***, including the feed and product streams, operating temperature, pressure, pH, retention time, bowl diameter, and rotational speed.

Table 7. Operating Conditions and Stream Information for the Decanter Centrifugation Units (C-101A/B/C/D).

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Pregnant Leaching Solution (PLS)
Product(s) out	Solids: Carbon (C), Zirconium oxide (ZrO_2), Tungsten (W), PVFD ($C_2H_2F_2$)
Temperature (°C)	50
Pressure (atm)	1
pH (if applicable)	<0
Retention Time	10-30 minutes [10]
Bowl Diameter (m)	0.52 [10]
Rotational Speed (rpm)	2,838 [10]

c. Reactions/Kinetics

Centrifugation is a physical separation process that does not involve chemical reactions or reaction kinetics, as separation occurs through centrifugal force and differences in density.

d. Performance / Separation Efficiencies

Table 8 below summarizes the separation efficiencies assumed for the decanter centrifuge during solid-liquid separation.

Table 8. Solid-Liquid Separation and Washing Efficiencies for the Decanter Centrifugation Unit. [10] [11]

Element/Component	**Efficiency (%)**
Pregnant Leaching Solution (PLS)	99.98
Solid Residue (Cake)	71.49
Repulping Liquid-to-Solid Ratio	3:1

e. Advantages & Disadvantages

- **Advantages:**

1. Continuous operation with high throughput, and continuous discharge of the solid cake. [7]

2. Effective on fine particles that settle or filter poorly. [12]

- **Disadvantages:**

1. The density difference between graphite (about 2.2 g/cm³) and the leachate (about 1.1–1.2 g/cm³) is modest, so very fine particles may remain in the centrate. A flocculant or a polishing step (membrane filtration in this process) may be required. [7] [9]
2. The acidic leachate demands corrosion-resistant materials such as duplex stainless steel, Hastelloy or lined components, which increases cost. [7]

A practical arrangement is therefore a decanter centrifuge for the bulk separation, followed by membrane filtration as the polishing step, as in the process described here.

f. Equipment Design

The primary solid-liquid separation and washing circuit is modeled in Aspen Plus using a 4-stage Counter-Current Decantation configuration using four LW520 Decanter Centrifuges paired with four repulping tanks operating in series.

1. Aspen Plus Block Selection & Flowsheet Configuration

- Solid-Liquid Separation Modeling: Solid-liquid separation is implemented using four sequential Sep (or Solid Separator) blocks. Each decanter centrifuge block receives the slurry feed and separates it into a clarified liquid centrate, and a concentrated wet cake based on user-defined split fractions.
- Intermediate Repulping Tanks: The wet solid cake discharged from each centrifuge is directed into a repulping tank (modeled as an Aspen Mixer block), where it is mixed with reverse-flowing wash liquor at a liquid-to-solid (L:S) mass ratio of 3:1 to 5:1 for a 10–30-minute retention time. [10]

2. Process Efficiency Parameters & Aspen Split Specifications

To reflect physical equipment performance and mass balance targets in Aspen Plus, the split fractions across the 4-stage decanter circuit are specified as follows:

- The decanter centrifuge achieves 99.98% bulk solid capture efficiency (0.9998 split to cake) for insoluble residue (primarily graphite, carbon black, PVDF binder, and solid metal fragments). [9]
- Only 0.02% of fine solids (0.0002 split to centrate) remain suspended in the clarified Pregnant Leach Solution (PLS) stream. [9]
- The centrifuge cake is discharged continuously with a solid residue efficiency of 71.49% by mass relative to the total wet cake weight. [9]
- The remaining 28.51% of the cake mass consists of entrained liquid (PLS) held within the porous void space. [9]
- In Aspen Plus, each repulping mixing stage is specified with a 75% single-stage washing efficiency or 3:1. [10]
- This stage of efficiency accounts for finite contact time (10–30 minutes) and liquid equilibrium dilution during repulping. [10]
- While a single stage removes 75% of the entrained metals, cascading four counter-current repulping stages yields an exponential increase in total recovery. [10]

$$\eta_{(circuit)} = 1 - (1 - 0.75)^4 = 1 - (0.25)^4 = 99.61 \qquad \textbf{[Equation 5]}$$

3. Sizing & Mechanical Specifications for LW520 Units

The four centrifuge units are specified based on standard industrial decanter dimensions to handle the continuous throughput of NMC111 black mass leach slurry:

- Equipment Model: LW520 Industrial Decanter Centrifuge (4 units operating in series).
- Bowl Diameter: 0.52 m (520 mm).
- Rotational Speed: 2,838 rpm, producing centrifugal acceleration exceeding 2,300 x g to ensure complete separation of fine graphite particles.
- Operating Conditions: 50 °C, 1 atm, and acidic media (pH < 0 to 2.0). [10]

g. Waste Treatment & Process Concerns

The solid cake discharged from the primary decanter centrifuge (C-101A) is wet, acidic, and entrained with Pregnant Leach Solution (PLS) held within its porous structure rich in (Ni^{2+}), (Co^{2+}), (Mn^{2+}), and (Li^{+}). To prevent these metals from leaving with the graphite-rich residue, the cake is washed counter-currently with dilute sulfuric acid (H_2SO_4) in water at a pH of roughly 1.5-2.0, matching the acidic conditions of the upstream Chemical Leaching Reactor (R-101). [9]

Washing displaces the entrained metal-rich liquid so it can be returned to the primary PLS stream. Maintaining an acidic wash medium (pH 1.5-2.0) is critical; using wash water with a higher or neutral pH would raise the localized pH within the cake, causing iron (Fe^{3+}), aluminum (Al^{3+}), and copper (Cu^{2+}) to hydrolyze and precipitate back onto the solids, trapping target metals and re-contaminating the graphite residue.

Additionally, washing lowers residual acidity over successive stages, making the final cake safer to handle, easier to neutralize downstream, and better aligned with processing requirements. Two distinct mechanisms of metal loss are addressed in this circuit:

- Entrained Solute Loss: Free liquid held within interstitial void spaces, which is readily removed by fluid displacement.
- Adsorbed Metal Loss: Metal cations weakly bound to the high-surface-area graphite matrix, requiring a slightly acidic wash and extended contact time to fully desorb.

The selected washing method is stirred-tank repulping and reseparation. Discharged wet cake drops into an agitated tank (T-101A) where it is repulped with wash liquid at a liquid-to-solid (L:S) mass ratio of 3:1 to 5:1. The slurry is mixed for **1**0 to 30 minutes to break up agglomerated solids and allow trapped metals to diffuse completely out of the microporous cake into the bulk liquid. The resulting slurry is then sent to a secondary decanter centrifuge to separate the washed cake from the metal-bearing wash liquor. Mechanical repulping provides significantly higher contact time and mass-transfer efficiency compared to other mechanisms such as spraying liquid over a moving cake inside a decanter centrifuge.

To maximize metal recovery while minimizing freshwater consumption, washing is executed counter-currently across 4 stages (T-101) paired with (C-101):

- Solids Path: The graphite cake moves forward sequentially from Stage 1 to Stage 4.
- Wash Liquid Path: Fresh dilute sulfuric acid wash liquid enters exclusively at the final stage (Stage 4 / T-101D) and is pumped backward from Stage 4 to Stage 1.
- Progressive Cleaning: At each stage, the solid cake meets a liquid wash stream that is cleaner than the stream in the preceding stage, driving higher solute extraction.

- PLS Integration: The wash liquor exiting Stage 1 is the most concentrated in desorbed metals and is piped directly back into the primary PLS stream (Stream 06) to ensure complete recovery.

  Under this configuration, the dirtiest cake meets the most concentrated wash liquid at the start, while the cleanest cake meets the freshest acid wash at the end. While a single repulping stage achieves a 75% single-stage washing efficiency, cascading four counter-current repulping stages yields a cumulative metal recovery efficiency of approximately 99%.

4. Copper Sulfide Precipitation Units

a. *Purpose*

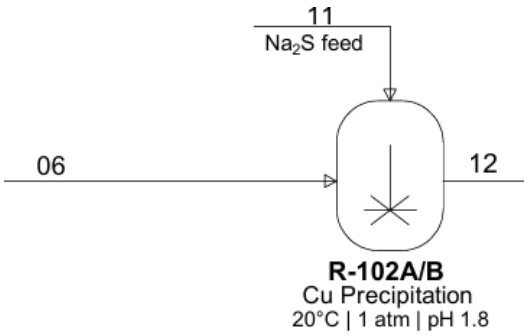


Figure 7. Process Flow Diagram of the Copper Sulfide Precipitation Unit (R-102A/B).

Copper sulfide precipitation is a chemical separation technique that uses sodium sulfide (Na_2S) to selectively remove dissolved copper from the metal-rich leach solution. Sodium sulfide supplies sulfide species that react with dissolved copper ions to form insoluble copper sulfide (CuS). The resulting slurry undergoes solid-liquid separation to recover a CuS filter cake and a copper-depleted filtrate. Process performance depends on temperature, pH, reagent concentration, reaction time, and mixing intensity. [13]

b. *Conditions*

Table 9 below summarizes the operating conditions used for the copper sulfide precipitation process.

Table 9. Operating Conditions and Stream Information for the Copper Sulfide Precipitation Units (R-102A/B). [14]

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Metal-rich leach solution; sodium sulfide stock solution.
Product(s) out	CuS filter cake; copper-depleted filtrate containing Li, Ni, Co, Mn, Fe and Al.
Temperature (°C)	20
Pressure (atm)	1
Operating pH	1.8
Na_2S Concentration (M)	0.20
Reaction Time (h)	Undetermined
Rotational Speed (rpm)	Undetermined

c. *Reactions/Kinetics*

Sodium sulphide dissociates in water:

1) $Na_2S \rightleftharpoons 2Na^+ + S^{2-}$ **[Equation 6]**

The dissolved sulphide is protonated:

2) $S^{2-} + H^+ \rightarrow HS^-$ **[Equation 7]**
3) $HS^- + H^+ \rightarrow H_2S$ **[Equation 8]**

Dissolved hydrogen sulphide can exceed solubility and escape as a gas:

4) $H_2S(aq) \rightarrow H_2S(g)$ **[Equation 9]**

Dissolved copper reacts with available sulphide to precipitate:

5) $Cu^{2+} + S^{2-} \rightarrow CuS(s)$ **[Equation 10]**
6) $Cu^{2+} + HS^- \rightarrow CuS(s) + H^+$ **[Equation 11]**
7) $Cu^{2+} + H_2S \rightarrow CuS(s) + 2H^+$ **[Equation 12]** [13]

d. Separation Efficiencies

Table 10 summarizes the copper precipitation and recovery efficiencies for the copper sulfide precipitation process.

Table 10. Copper Precipitation and Recovery Efficiencies. [14]

Element/Component	Efficiency (%)
Copper precipitated	96.19
Copper recovered in CuS cake	96.00

e. Alternatives

LIX 984 solvent extraction

Advantages: Alternative copper separation route without adding sulfide.

Disadvantages: Requires organic-phase handling, extraction and stripping equipment, and reagent recycling. Lower separation efficiency.

f. Waste Treatment & Process Concerns

Safety

- Potential H_2S formation requires gas management during sulfide addition and subsequent acidification. [13] [15]
- Excess sulfide can cause loss of valuable metals up to 4% and carry over to downstream recovery. [14]
- The CuS cake requires characterization before copper recovery or waste disposal.

Safety Measures

1) Equip Provide controlled reagent dosing, pH monitoring and appropriate H_2S detection and off-gas treatment. [15]

2) Verify residual sulfide before downstream processing and prevent uncontrolled acidification of sulfide-bearing streams.

3) Use enclosed solids handling and collect filtrate and wash solutions for treatment or controlled recycle.

4. Nitrogen Sparging/H_2S Removal Unit

a. Purpose

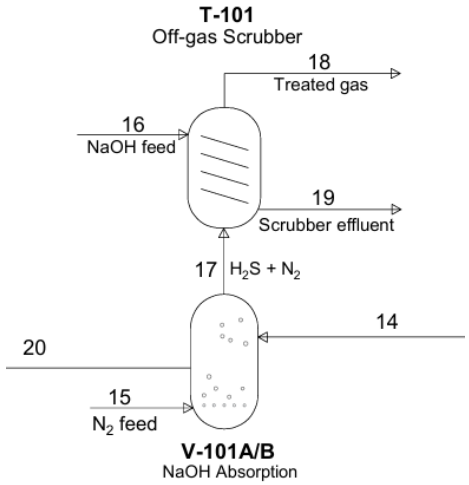


Figure 8. Process Flow Diagram of the Nitrogen Sparging Unit (V-101A/B).

To remove residual dissolved hydrogen sulfide from the filtered copper-precipitation liquor before iron and aluminum precipitation. Nitrogen is contacted with the liquid in an enclosed vessel or column, carrying H_2S into a separate gas stream for treatment. Nitrogen acts as a striping gas and is not consumed in a chemical reaction. Continuous nitrogen contact provides fresh gas with a low H_2S partial pressure and additional gas–liquid contact area. Natural release and deliberate stripping must both feed a contained gas-treatment system. [16]

b. Conditions

Table 11 below summarizes the operating conditions used for the H_2S removal process.

Table 11. Operating Conditions and Stream Information for the Nitrogen Sparging Unit (V-101A/B). [16]

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Filtered copper-stage liquor; nitrogen gas
Product(s) out	Metal-bearing solution with reduced dissolved sulfide; nitrogen, H_2S gas and water vapour
Temperature (°C)	25
Pressure (atm)	1
Feed pH	1.8 ± 0.1

c. Reactions/Kinetics

Sparging is a physical separation process that only involves mass transfer separations as it occurs through differences in the partial pressure of H_2S and pure nitrogen gas. [16]

d. Separation Efficiencies

Table 12 below summarizes the separation efficiency of H_2S during the sparging process.

Table 12. Hydrogen Sulfide Removal Efficiencies for the Nitrogen Sparging Process.

Element/Component	**Efficiency (%)**
After venting	50
After sparging	95
Total removal	97.5

e. Alternatives

1. Using hydrogen peroxide oxidation

Advantages: Chemically converts residual sulfide into elemental sulfur or sulfate, reducing the amount of H_2S requiring gas treatment.

Disadvantages: Requires additional chemicals and careful dosing. It may produce sulfur solids, change iron's oxidation state, and dissolve remaining CuS particles, returning copper to the solution. Treatment should therefore be evaluated after CuS filtration. [17]

2. Using steam stripping

Advantages: Uses steam to transfer dissolved H_2S from the solution into an overhead vapor stream. It is an established industrial treatment method and may be attractive when inexpensive steam or waste heat is available.

Disadvantages: Requires heating a large volume of solution, increasing energy consumption. The H_2S-containing overhead still requires treatment, and volatile solvents may also be stripped. Additional equipment and cooling may be needed before downstream precipitation. [18]

3. Using vacuum stripping

Advantages: Uses reduced pressure to encourage dissolved H_2S to leave the solution without requiring steam or additional stripping gases. Can operate at relatively low temperatures, reducing energy consumption associated with heating. Does not introduce oxygen or additional chemicals into the solution help preserve the metal concentrations and acidic conditions.

Disadvantages: Requires specialized vacuum equipment and sealed vessels, increasing capital and maintenance costs. The H_2S-containing gas still requires treatment before release. Vacuum

operation may also cause water evaporation, foaming, and the removal of other volatile compounds. Removal efficiency depends on operating pressure, temperature, and contact time. [19]

f. Waste Treatment & Process Concerns

Safety

1. Hydrogen sulfide is highly toxic and flammable. Its smell cannot be relied upon because exposure can reduce the ability to detect it. [15]
2. Nitrogen leaks can displace oxygen and create an asphyxiation hazard, particularly in enclosed areas.
3. Venting and sparging transfer H_2S from the liquid into the gas stream. Failure of the collection or scrubbing system could release this gas into the workplace.
4. The acidic precipitation solution and caustic scrubber liquid present splash and corrosion hazards. [22]
5. Spent caustic contains dissolved sulfides such as NaHS. Contact with acidic waste can release H_2S again; these waste streams must remain segregated.
6. The removal percentages do not establish a safe outlet concentration. Sulfide remaining in the liquid must be considered in downstream processing.

Safety Measures

1. Connect the reactor vent and sparging exhaust to a dedicated scrubber rather than releasing untreated gas.

2. Provide fixed H_2S detectors, appropriate personal monitors and oxygen-deficiency monitoring where nitrogen could accumulate. [15]
3. Establish an engineered shutdown sequence that maintains containment and treatment.
4. Measure scrubbing-liquid pH, circulation flow, pressure drop and outlet-gas H_2S. Replace or treat spent solution before absorption performance becomes inadequate.
5. Measure residual dissolved sulfide before transferring to the next precipitation stage. Use measured concentrations to validate the assumed stripping efficiency.
6. Provide a demister, suitable pressure/vacuum protection and materials selected for the actual acid, sulfide and caustic conditions.
7. Include appropriate gloves, goggles, face protection, eyewash and emergency showers. Establish evacuation and confined-space procedures; rescue must be carried out by trained, equipped personnel

5. Iron and Aluminium Precipitation Units

a. Purpose

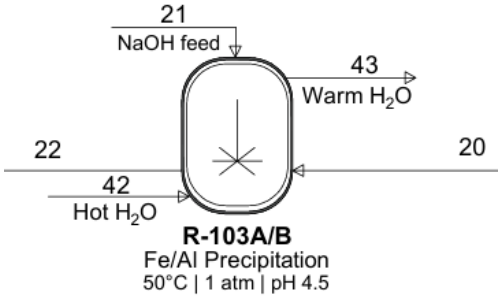


Figure 9. Process Flow Diagram of the Iron and Aluminium Precipitation Unit (R-103A/B).

Iron and aluminum precipitation is a chemical separation technique that uses controlled NaOH dosing to selectively recover iron and aluminum from the copper deficient leach solution. The unit is described as an agitated precipitation reactor. Sodium hydroxide is used to increase the pH and lower solubility to form a mixed hydroxide precipitate. The resulting mixture is then subjected to solid-liquid separation through vacuum filtration, producing an aluminum and iron rich cake and a filtrate at a pH of 4.5 and deficient of all impurities. The performance process depends on operating conditions such as temperature, pH, reagent concentration, reaction time, and mixing intensity. [20]

b. Conditions

Table 13 below summarizes the operating conditions used for the Iron and Aluminum precipitation process.

Table 13. Operating Conditions and Stream Information for the Iron and Aluminum Precipitation Unit (R-103A/B).

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Cu-depleted, sulfide-controlled liquor; NaOH solution.
Product(s) out	Fe/Al-rich filter cake and filtrate containing Ni, Co, Mn and Li.
Temperature (°C)	60
Pressure (atm)	1
Feed pH	2 ± 0.1
Reaction Time (h)	6
Rotational Speed (rpm)	620

c. Reactions/Kinetics

Sodium hydroxide disassociates in water and neutralizes acid:

1) $NaOH \rightarrow Na^{+} + OH^{-}$ **[Equation 13]**

2) $OH^{-} + H^{+} \rightarrow H_2O$ **[Equation 14]**

At a higher pH ferric and aluminium ions hydrolyze and precipitate as a hydroxide:

3) $Fe^{3+} + 3OH^{-} \rightarrow Fe(OH)_{3(s)}$ **[Equation 15]**

4) $Al^{3+} + 3OH^{-} \rightarrow Al(OH)_{3(s)}$ **[Equation 16]** [21]

d. Separation Efficiencies

Table 14 below summarizes the precipitation efficiencies of aluminium and iron in the precipitation process.

Table 14. Iron and Aluminum Precipitation Efficiencies.

Element/Component	**Efficiency (%)**
Fe^{3+} and Al^{3+} removal by hydroxide precipitation	Approximately 99% at pH 4
NMC Precipitation	4-5
Filter capture	98.7

e. Alternatives

1. Using Phosphate precipitation

Advantages: a similar study achieved higher recovery at a lower pH and with a faster precipitation.

Disadvantages: Requires additional phosphorus-containing reagent and will overcomplicate downstream recovery with residual phosphate. [21]

2. Using Lithium hydroxide

Advantages: Avoids adding sodium as lithium will be part of downstream removal anyways. [20]

Disadvantages: More expensive and the potential loss of added lithium in the cake.

- *Waste Treatment & Process Concerns*

Safety

1. Sodium hydroxide is corrosive, and acid neutralization releases heat. Rapid addition can cause local overheating and splashing. [22]
2. Residual upstream peroxide may decompose in contact with metal-containing solution, generating oxygen and heat. [23]
3. Poor pH control can increase valuable-metal losses to the impurity cake.
4. The filter cake contains metals and retained process solution and requires characterization before disposal or recovery

Safety Measures

1. Use controlled caustic dosing with continuous pH and temperature monitoring.
2. Provide mixing during reagent addition and interlocks to stop dosing following loss of agitation or excessive temperature.
3. Verify residual sulfide and peroxide before transferring the feed into this unit; retain controlled off-gas collection where required.
4. Provide chemical-resistant gloves, eye and face protection, eyewash facilities, and emergency showers.
5. Select construction materials for the actual acid, caustic, fluoride, and temperature conditions.

6. NMC Oxalate Precipitation Units

a. Purpose

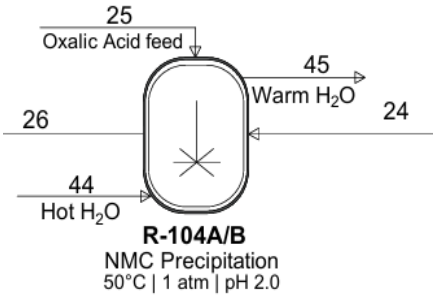


Figure 10. Process Flow Diagram of the NMC Oxalate Precipitation Unit (R-104A/B).

NMC oxalate precipitation is a chemical separation technique that uses oxalic acid to selectively recover nickel, cobalt, and manganese from the metal-rich leach solution. The unit is described as an agitated precipitation reactor. Oxalic acid reacts with the dissolved metal ions to form insoluble metal oxalates, which precipitate as suspended solids. The resulting mixture is then subjected to solid-liquid separation through vacuum filtration, producing an NMC oxalate-rich filter cake and a lithium-rich filtrate. The performance process depends on operating conditions such as temperature, pH, reagent concentration, reaction time, and mixing intensity.

b. Conditions

Table 15 below summarizes the operating conditions used for the NMC oxalate precipitation process.

Table 15. Operating Conditions and Stream Information for the NMC Oxalate Precipitation Unit (R-104A/B). [24] [26]

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Purified metal-rich slurry containing Nickel, Cobalt, Manganese, Lithium, and residual impurities; Oxalic acid stock solution
Product(s) out	Slurry containing precipitated nickel, cobalt, and manganese oxalates, with lithium and remaining impurities in the aqueous phase
Temperature (°C)	50
Pressure (atm)	1
Feed pH	2 ± 0.1
Acid-to-Metal (A/M) Ratio	Determined by ICP-OES and leaching chemistry
Reaction Time (h)	3
Rotational Speed (rpm)	750

c. Reactions/Kinetics

Oxalic acid dissociates in water:

1) $H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^-$ **[Equation 17]**
2) $HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2-}$ **[Equation 18]**

The oxalate then forms solid metal oxalates:

3) $Ni_2^+ + C_2O_4^{2-} \rightarrow NiC_2O_{4(s)}$ **[Equation 19]**
4) $CO_2 + C_2O_4 \rightarrow CoC_2O_{4(s)}$ **[Equation 20]**
5) $Mn^{2+} + C_2O_4^{2-} \rightarrow MnC_2O_{4(s)}$ **[Equation 21]** [24] [25]

d. Separation Efficiencies

Table 16 below summarizes the precipitation efficiencies of nickel, cobalt, and manganese during the oxalate precipitation process.

Table 16. NMC Oxalate Precipitation and Separation Efficiencies. [3] [25] [34]

Element/Component	**Efficiency (%)**
Ni oxalate (NiC_2O_4)	98.1
Cobalt oxalate (CoC_2O_4)	92.0
Manganese oxalate (MnC_2O_4)	98.7
NMC Precipitation + Filtration (NMC)	98.55, 86.87, 99.22

e. Alternatives

1. Using sulfide precipitation (chemical or biogenic)

Advantages: Consistently outperforms recovery percent wise.

Disadvantages: H_2S becomes a much larger issue, and the biogenic route is not selective at all. Oxalate is more selective and allows use to only remove NMC solids while leaving the lithium to be treated elsewhere. [13]

2. Using Ammonium oxalate or Sodium oxalate.

Advantages: Reduces safety risk and helps keep the feed requirements much less strict.

Disadvantages: More expensive than oxalic acid, and the leftover ion may become a contaminant depending on separation.

f. Equipment Design

Vacuum filtration was selected for separating precipitated Ni, Co, and Mn oxalate solids from the Li-rich aqueous solution. A polyether sulfone (PES) membrane with a pore size of 0.45 μm was selected based on the experimental filtration of mixed NMC oxalate precipitates reported by Do et al. (2022). For the rotary drum filtration unit, the required filtration area will be determined based on processing capacity, cake permeability, and filtration resistance. The solid-liquid separation was modelled in Aspen Plus using a SEP2 block, which represents the distribution of components between the wet filter cake and filtrate based on specified separation parameters. Preliminary filtration recoveries of 98.55%, 99.22%, and 86.87% were considered for NiC_2O_4, CoC_2O_4, and MnC_2O_4, respectively, while a wet cake moisture content of 20 wt% was assumed for the material balance. [25]

g. Waste Treatment & Process Concerns

Safety

1. If any Na_2S remains following the purification step, it can make H_2S which is toxic and highly flammable. [15]
2. If any phosphate remains following the purification step, it could react with the transition metals in solution before the oxalate, effectively wasting product and fouling filters downstream. [21]
3. If any hydroxide remains following the purification step, it could react with the acid, wasting it to create water, and changing the reactor pH.

4. The pH is a very important condition we need to control in the reactor for the reactions to take place; it will be very low so it will be harmful to encounter. [20] [24]
5. Oxalic acid itself is a hazard to handle, being toxic and corrosive to the eyes and skin.
6. There is a risk of corrosion in the reactor and piping.

Safety Measures

1. Equip the facility with hydrogen sulfide detectors and alarm systems to identify potential gas leaks.
2. Monitor phosphate concentrations as an impurity and verify that they remain below the established threshold.
3. Establish a pH acceptance criterion that must be satisfied before transferring the slurry into the precipitation reactor.
4. Provide appropriate personal protective equipment (PPE), accessible eyewash stations, and spill response kits for emergency containment and neutralization.
5. Ensure adequate ventilation and the use of appropriate PPE to minimize chemical exposure.
6. Construct the reactor and piping using glass-lined materials or high-grade stainless steel to reduce corrosion risks.

7. Vacuum Filtration Units

a. Purpose

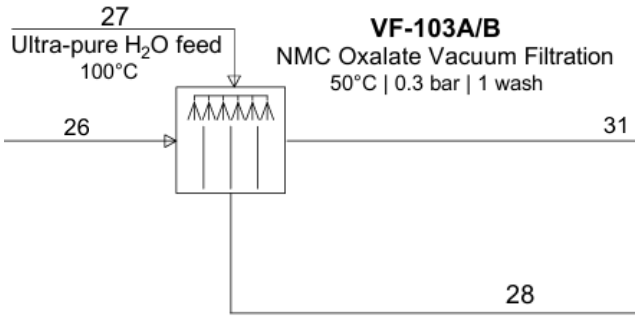


Figure 11. Process Flow Diagram of the NMC Vacuum Filtration Unit (VF-101A/B).

The purpose of the NMC Rotary Vacuum Drum Filtration Unit (VF-101A/B) is to separate the slurry received from the NMC oxalate precipitation unit into a wet filter cake containing the recovered Ni, Co, and Mn oxalates and a Li-rich filtrate directed downstream for further processing. Based on preliminary research, rotary vacuum drum filters can operate at pressure differences of approximately 85 kPa, making them suitable for the proposed solid-liquid separation. Additionally, the unit incorporates a cake washing system equipped with spray nozzles to remove residual soluble impurities and improve the purity of the recovered NMC oxalates. [29] [30] [31]

b. Conditions

Table 17 below summarizes the preliminary operating conditions selected for the NMC vacuum filtration process. These conditions were established using laboratory-scale literature and engineering assumptions based on the proposed separation process. The operating parameters will

be further evaluated and optimized during subsequent design stages and should therefore be considered preliminary values for review.

Table 17. Operating Conditions and Stream Information for the NMC Vacuum Filtration Unit (VF-101A/B).

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Slurry from the NMC oxalate precipitation reactor containing suspended Ni, Co, and Mn oxalate solids, an aqueous solution dissolved in Li, residual oxalic acid/oxalate, and remaining impurities. [25]
Product(s) out	Wet NMC filter cake sent to drying process and Li-rich filtrate containing dissolved impurities. [25]
Feed Pressure (atm)	1.0 (atmospheric feed pressure).
Feed Temperature (°C)	50
pH	Adopted from upstream NMC oxalate precipitator.
Wet Cake Liquid Content	30 wt% (preliminary modelling assumption).
Vacuum Pressure Difference, ΔP (atm)	~0.69 [30]
Filter Medium	Polyether sulfone (PES) membrane. [32]
Membrane Pore size (μm)	0.45 [32]
Filter Area (m^2)	To be determined through equipment sizing.
Cake Washing	Single wash with 100ºC ultra-pure water. [31] [35]
Wash Efficiency	98% of removing soluble impurities during washing. [31]
Water-to-solids Ratio (kg/kg)	1:1 (preliminary modelling assumption).
Drum size	Further research to be done.
Rotational Speed (rpm)	Further research to be done.

c. Reactions/Kinetics

Vacuum Filtration is a physical separation process that does not involve chemical reactions or reaction kinetics. Solid-liquid separation occurs through an applied pressure difference, with suspended solids retained by the filter medium and the liquid passing through as a filtrate.

d. Separation Efficiencies

Table 18 below summarizes the combined recovery efficiencies for the NMC oxalate precipitation and vacuum filtration processes, including the recovery of Ni, Co, and Mn oxalate solids from the Li-rich aqueous solution.

Table 18. Combined NMC Oxalate Precipitation and Filtration Recoveries. [34]

Element/Component	**Efficiency (%)**
Ni oxalate (NiC_2O_4)	98.55
Cobalt oxalate (CoC_2O_4)	99.22
Manganese oxalate (MnC_2O_4)	86.87

e. Advantages & Disadvantages

Centrifugation

- **Advantages**: Does not require filter media, is less dependent on cake permeability, and can be effective for solid-liquid separation for fine particles.
- **Disadvantages**: Higher energy consumption, greater mechanical complexity, increased maintenance requirements due to rotating components, and potentially higher capital costs.

Pressure Filtration

- **Advantages**: Operates at higher pressure differentials (ΔP), potentially increasing filtration rates and reducing residual cake moisture.

- **Disadvantages**: Requires pressure-rated equipment, more complex operating and maintenance procedures, and potentially higher capital and operating costs compared with vacuum filtration.

f. Equipment Design

Vacuum filtration was selected for separating precipitated Ni, Co, and Mn oxalate solids from the Li-rich aqueous solution. A polyether sulfone (PES) membrane with a pore size of 0.45 μm was selected based on the experimental filtration of mixed NMC oxalate precipitates reported by Do et al. (2022). For the rotary drum filtration unit, the required filtration area will be determined based on processing capacity, cake permeability, and filtration resistance. The solid-liquid separation was modelled in Aspen Plus using a SEP2 block, which represents the distribution of components between the wet filter cake and filtrate based on specified separation parameters. Preliminary filtration recoveries of 98.55%, 99.22%, and 86.87% were considered for NiC_2O_4, CoC_2O_4, and MnC_2O_4, respectively, while a wet cake moisture content of 20 wt% was assumed for the material balance.

g. Waste Treatment & Process Concerns

Safety

- Acidic slurry and residual oxalic acid may cause equipment corrosion and chemical exposure.
- Residual oxalate in the slurry and filtrate requires appropriate handling and disposal.

- Fine NMC oxalate particles may pass through an unsuitable filter medium, reducing solids recovery. [29]
- Accumulation of fine particles can clog the filter medium, reducing filtration rates and increasing maintenance requirements. [25] [35]
- Increasing filter cake thickness increases filtration resistance, potentially reducing throughput. [29]
- Li-rich solution retained in the wet filter cake represents lithium loss and may contaminate the recovered NMC product. [31] [35]
- Vacuum filtration equipment may experience mechanical failure if the components are not adequately rated for the operating pressure difference. [30]
- Operating challenges can occur when working with a reduced pressure which may cause flashing or evaporation if the liquid's vapour pressure approaches or exceeds the absolute pressure within the filtration unit. [36]

Waste

- The filter medium may become fouled or damaged over time, requiring replacement, and generating solid waste potentially contaminated with residual metal oxalates and process solution. [35]
- Filter cake washing generates additional wastewater containing dissolved lithium, residual oxalates, and other imputes, which must be treated or recycled appropriately. [31]

8. Lithium Carbonate Precipitation Units

a. Purpose

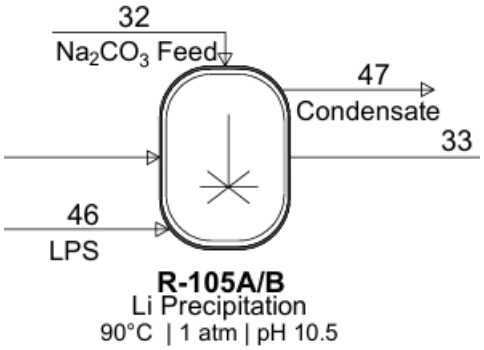


Figure 12. Process Flow Diagram of the Lithium Carbonate Precipitation Unit (R-105A/B).

The purpose of the Lithium Carbonate Precipitation Reactors (R-105A/B) is to recover dissolved lithium from the lithium-rich solution obtained following NMC oxalate filtration. Sodium carbonate (Na_2CO_3) is introduced as the precipitating agent, reacting with dissolved lithium ions to form solid lithium carbonate (Li_2CO_3). The precipitation process is conducted at approximately 90°C and atmospheric pressure, with the pH maintained near 10.5 to promote lithium carbonate formation. The resulting slurry, containing precipitated Li_2CO_3 crystals and the remaining aqueous solution, is directed downstream to the lithium carbonate filtration and washing units (F-103A/B) for solid-liquid separation and product purification. [4]

b. Conditions

Table 19 below summarizes the preliminary operating conditions selected for the NMC vacuum filtration process. These conditions were established using laboratory-scale literature and engineering assumptions based on the proposed separation process. The operating parameters will

be further evaluated and optimized during subsequent design stages and should therefore be considered preliminary values for review.

Table 19. Operating Conditions and Stream Information for the Lithium Carbonate Precipitation Unit (R-105A/B). [4] [27] [28]

Parameter	**Operating Conditions/Stream Information**
Feed(s) in	Lithium-rich aqueous filtrate containing residual impurities, Sodium carbonate solution, and Sodium hydroxide solution for pH adjustment.
Product(s) out	Slurry containing precipitated Lithium carbonate solids.
Temperature (°C)	90
Pressure (atm)	1
pH	10.5
Reaction Time (h)	1
Carbonate-to-Lithium Molar Ratio	Stochiometric excess: 1:1.1 (to drive precipitation).
Rotational Speed (rpm)	600

c. Reactions/Kinetics

Lithium ions combine with carbonate ions provided by sodium carbonate solution:

$2Li^{+} + CO_3^{2-} \rightarrow Li_2CO_{3(s)}$ **[Equation 22]** [4]

d. Separation Efficiencies

Precipitation and Filtration Efficiency: 77.97%, determined experimentally using ICP-OES analysis. [4]

e. Alternatives

Using Carbon Dioxide Instead of Sodium Carbonate

Advantages: Eliminates an additional source of sodium, reducing the burden on downstream evaporators during crystallization. Utilizes carbon dioxide waste and can achieve high recovery of high-purity lithium carbonate.

Disadvantages: Requires a pressurized reactor to improve carbon dioxide dissolution and gas-liquid mass transfer. Gas-liquid reactions are generally slower than liquid-liquid reactions. Additional base dosing is required because dissolved carbon dioxide forms carbonic acid, lowering the solution pH. The process is also more difficult to model. [27]

f. Waste Treatment & Process Concerns

Safety

1. Agitating a near-boiling, highly basic (approximately pH 11) slurry presents significant scalding and chemical burn hazards.
2. Rapid addition of sodium carbonate to neutralize the feed can cause excessive carbon dioxide generation, potentially resulting in slurry foaming and damage to seals or vents.

Safety Measures & Waste Management

The process generates highly saline wastewater that requires further treatment.

1. Insulate and double-jacket the reactor vessels and implement an automated sampling process to minimize exposure hazards.

2. Adjust feed pH to 7 using sodium hydroxide before adding sodium carbonate to reduce gas generation and reagent costs.

3. Evaporate the saline wastewater to crystallize dissolved salts into a solid cake for subsequent handling or disposal. [4] [27]

III. Project Statement

The project will develop a preliminary hydrometallurgical process to recover critical metals from spent NMC black mass. The process should define the major unit operations, material recovery sequence, feed basis, plant capacity, and a potential location in Canada.

The research estimates the black mass production basis from using Li-Cycle data: 35,000 tonnes/year of black mass from approximately 90,000 tonnes/year of LIB feed; corresponding to an estimated 38.9% conversion of LIB feed to black mass.

IV. Thermodynamics & Kinetics

Thermodynamics

Reactor R-101:

In batteries the valuable metals like cobalt, nickel, and manganese will be locked within a solid crystal oxide lattice. Sulfuric acid is not that strong that it can help break down this crystal structure to extract valuable metals. By the addition of hydrogen peroxide an electron donor is introduced. This electron will naturally jump from the hydrogen peroxide to high energy metals like cobalt and nickel which will reduce their electronic charge.

$Oxidation: H_2O_2 \rightarrow O_2 + 2H^+ + 2e^- \ (E^o = +0.695\ V)$ **[Equation 23]**

$Reduction: Co^{3+} + e^- \rightarrow Co^{2+} (E^o = +1.830\ V)$ **[Equation 24]**

As $E^o_{cell} > 0$, the Gibbs free energy change is negative. The redox transfer breaks the oxide lattice bonds and releases soluble $Co^{2+}, Ni^{2+}, Mn^{2+}, and\ Li^+$ into the aqueous sulfate solution.

Reactor R-102 & R-103:

Once the solution is dissolved it contains the target metals (Ni, Co, Mn, Li) mixed with unwanted impurities such as copper, iron, and aluminum. Every metal compound will have a solubility limit or K_{sp}. This defines the maximum ion concentration the solution can hold before solid precipitation occurs.

By adding sodium sulfide under strong acidic conditions, the copper solidifies so it can be filtered out while nickel and cobalt stay dissolved.

By adding sodium hydroxide to raise the pH to 4.2 will cause iron and aluminum to form solid hydroxide while nickel, cobalt, and manganese, and lithium remain liquid.

Reactor R-104:

To separate the nickel, cobalt, and manganese from lithium, oxalic acid is added while holding the pH at 2.0. The oxalic acid will break down to form negatively charged oxalate ions. These ions will lock together the nickel, cobalt, and manganese to build solid crystals that sink to the bottom. Lithium does not bond with the oxalate and stays dissolved in the liquid creating a clean separation.

Reactor R-105:

Lithium Carbonate (Li_2CO_3) becomes much less soluble as the solution gets hotter. By heating the liquid up to 90°C and adding sodium carbonate at a high pH of 10.5 the lithium will be forced out of the water a pure white solid.

<u>Kinetics</u>

Reactor R-101:

The dissolution of spent NMC cathode oxides ($LiNi_{\frac{1}{3}}Mn_{\frac{1}{3}}Co_{\frac{1}{3}}O_2$) in aqueous sulfuric acid (H_2SO_4) with hydrogen peroxide (H_2O_2) as a reducing agent is a solid-liquid reaction. The overall reaction is controlled by three steps:

1. $H^+ \; and \; H_2O_2$ travel from the liquid film around the particle where liquid boundary diffusion happens.

2. Reagents penetrate through the outer porous layer which leads to porous diffusion.
3. The redox reaction breaks the metal-oxygen ionic bonds at the surface.

Lithium, Nickel, and Manganese exhibit rapid kinetics, reaching 100% dissolution within 15 minutes. Cobalt has slower reaction kinetics as it achieves 93% yield at 15 minutes and requires a full 60 minutes to reach 100% dissolution. As explained above, adding the full volume of hydrogen peroxide using the INIT Method faster initial reaction kinetics than gradual staged additions, the MULT method.

Reactor R-104:

The precipitation of nickel, cobalt, and manganese oxalates ($MC_2O_4 * 2H_2O$) has two main kinetic phenomena that need to be examined:

1. Primary Nucleation: In which there is a rapid birth of microscopic solids see crystals from supersaturated solutions.
2. Crystal Growth: Dissolved metal and oxalate ions depositing onto existing particle surfaces.

Rate Controlling Factors include the control of the pH which regulates the kinetic dissociation rate of oxalic acid into oxalate ions ($C_2O_4^{2-}$). High agitation eliminates the mass transfer limitations in the fluid ensuring uniform contact between the oxalic acid and metal ions while controlling the crystal size distribution.

The reactor requires a 3-hour retention time to allow sufficient time for crystal growth and reaches high equilibrium co-precipitation yields.

Reactor R-105:

Lithium Carbonate Precipitation Kinetics by combining dissolved lithium ions with carbonate ions to generate solid lithium carbonate:

$$2Li^{+}_{(aq)} + {CO_3}^{2-}_{(aq)} \rightarrow Li_2CO_{3(s)}$$ **[Equation 25]**

Rate Controlling Factors:

Temperature: Higher temperatures drive the reaction kinetics as they accelerate mass transfer and lowers solubility, creating the kinetic driving force for crystallization.

Agitation: Keeps the crystal nuclei suspended in solution to promote growth over nucleation resulting in larger particles.

The crystallization kinetics are relatively fast compared to oxalate precipitation reaching near-equilibrium solid yields within a 1-hour retention time.

V. Process Flow Diagram

The Process Flow Diagram (PFD) was developed using AutoCAD to illustrate the proposed hydrometallurgical recovery process for NMC111 lithium-ion battery black mass. The diagram identifies the major unit operations, equipment designations, operating conditions, and numbered process and service streams. It illustrates the sequence of chemical leaching, solid-liquid separation, impurity removal, selective precipitation, filtration, washing, and drying operations required to recover nickel, cobalt, and manganese as mixed metal oxalates and lithium as lithium carbonate (Li_2CO_3). The PFD also establishes a reference for conducting material and energy balances and evaluating the individual unit operations throughout the proposed process.

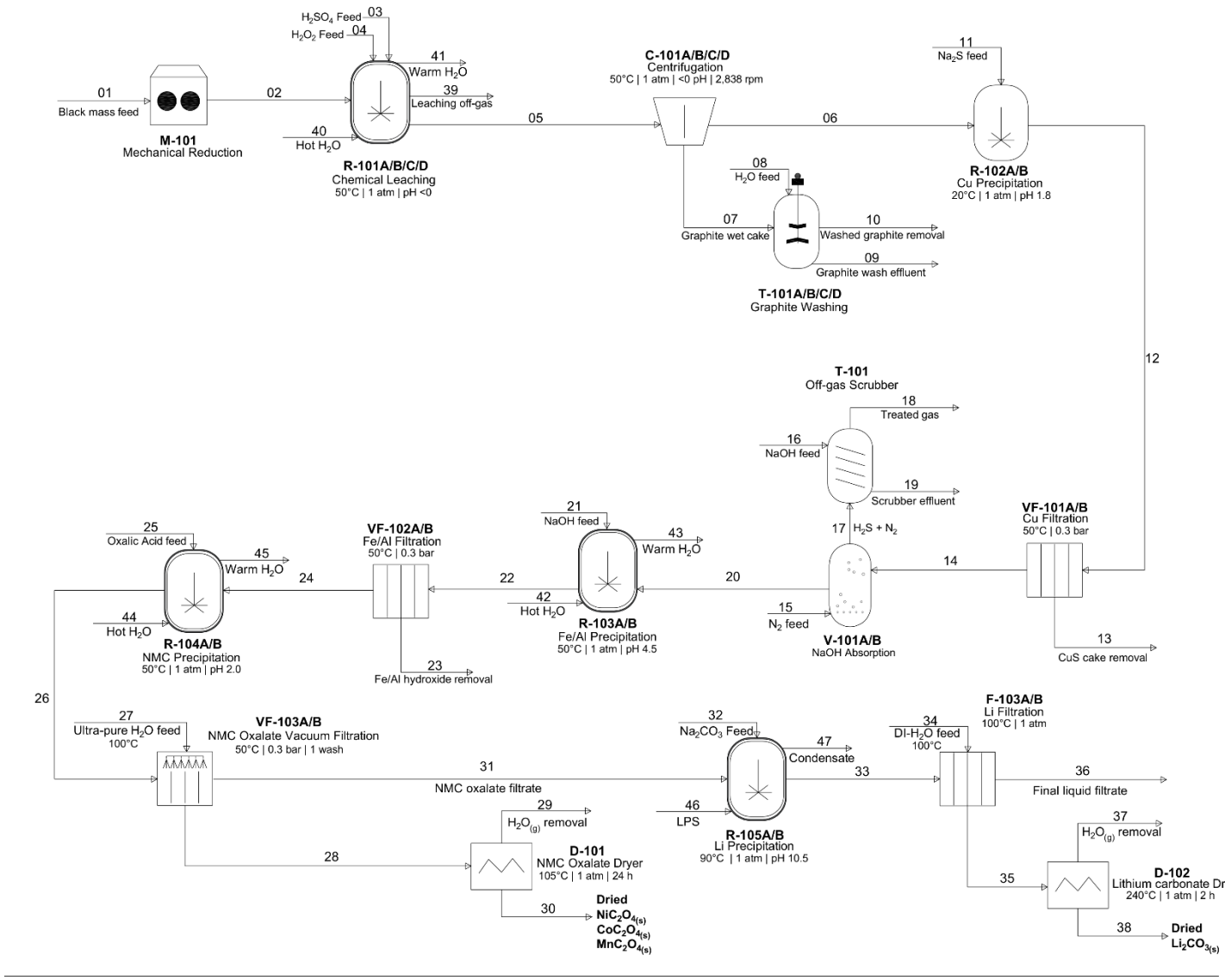


Figure 13. Process Flow Diagram of the Hydrometallurgical Process for NMC 111 Black Mass (Reproduced from Figure 1)

VI. Mass & Energy Balances

Mass Balance

Table 20 summarizes the 39 process streams, including their descriptions, major components, and total mass flow rates. The process begins with a black mass feed of 35,000 ton/yr, which undergoes chemical leaching using 837,369 ton/yr of sulfuric acid solution and 26,102 ton/yr of hydrogen peroxide solution. Following graphite separation, copper removal, and Fe/Al precipitation, the process produces **28,250 ton/yr of dried Ni/Co/Mn oxalates and 5,558 ton/yr of dried lithium carbonate product**. The table also highlights the significant liquid flows throughout the process, particularly the final lithium filtration stream of 3,983,410 ton/yr.

Table 20. Summary of Process Stream Descriptions, Compositions, and Total Mass Flow Rates

Stream	Process Section	Stream Description	Total Flow (ton/yr)
1	Black Mass Feed Preparation	Black mass feed	35,000
2	Black Mass Feed Preparation	Mechanically reduced NMC111 black mass	35,000
3	Chemical Leaching	H_2SO_4 feed	837,369
4	Chemical Leaching	H_2O_2 feed	26,102
5	Chemical Leaching	Leaching slurry to decanter	892,046
6	Graphite Separation / Copper Precipitation	Metal-rich leach solution	879,075
7	Graphite Separation	Graphite-rich wet cake	12,971
8	Graphite Washing	Graphite washing water	12,531
9	Graphite Washing	Graphite wash effluent	7,601

10	Graphite Washing	Washed graphite	17,901
11	Copper Precipitation	Na_2S feed	40,389
12	Copper Precipitation / Filtration	CuS slurry	919,598
13	Copper Filtration	CuS filter cake	522
14	Copper Filtration / Nitrogen Sparging	Copper-depleted filtrate	919,077
15	Nitrogen Sparging	N_2 feed	15,590
16	Off-Gas Scrubbing	NaOH feed	2,458
17	Nitrogen Sparging / Off-Gas Scrubbing	H_2S and N_2 off-gas	15,671
18	Off-Gas Scrubbing	Treated off-gas	15,590
19	Off-Gas Scrubbing	Scrubber effluent	2,539
20	Nitrogen Sparging / Fe-Al Precipitation	Sparged leach solution	918,996
21	Fe/Al Precipitation / Filtration	NaOH feed	2,290,321
22	Fe/Al Precipitation / Filtration	Fe/Al hydroxide slurry	3,209,317
23	Fe/Al Filtration	Fe/Al hydroxide solids	1,607
24	Fe/Al Filtration / NMC Precipitation	Purified filtrate	3,207,710
25	NMC Precipitation	Oxalic acid feed	220,819
26	NMC Precipitation / Vacuum Filtration	Mixed NMC oxalate slurry	3,455,034
27	NMC Oxalate Vacuum Filtration	Ultrapure washing water	28,247
28	NMC Oxalate Vacuum Filtration / Drying	Wet NMC oxalate cake	40,357
29	NMC Oxalate Drying	Water vapour	12,107

30	NMC Oxalate Drying	Dried NMC oxalate product	28,250
31	NMC Oxalate Vacuum Filtration / Li Precipitation	Lithium-rich filtrate	3,442,924
32	Lithium Precipitation	Na_2CO_3 feed	543,054
33	Lithium Precipitation / Filtration	Lithium carbonate slurry	3,985,978
34	Lithium Filtration	Deionized washing water	5,371
35	Lithium Filtration / Drying	Wet lithium carbonate filter cake	7,939
36	Lithium Filtration	Final liquid filtrate	3,983,410
37	Lithium Carbonate Drying	Water vapour	2,382
38	Lithium Carbonate Drying	Dried Li_2CO_3 product	5,558
39	Chemical Leaching	Leaching off-gas	3,440

Assumptions

The material balance calculations were developed using the researched elemental composition of NMC111 black mass from Canada and the separation efficiencies specified for each unit operation. The feed composition was expressed in weight percentages, while the efficiencies for leaching, centrifugation, copper precipitation and removal, Fe/Al precipitation and removal, NMC oxalate precipitation, vacuum filtration, lithium carbonate precipitation, and drying were applied to determine the distribution of materials throughout the process. Where experimental data were unavailable, engineering assumptions were introduced to estimate reaction conversions, reagent requirements, liquid retention, and product recovery. Several separation stages were assumed to

operate at 100% efficiency for selected components, representing idealized separation conditions. The density of water at 50°C was used where required for volumetric calculations.

For the chemical leaching stage, a 50% conversion of $LiPF_6$ through hydrolysis was assumed due to limited available experimental data. A Fe^{3+} to Fe^{2+} ratio of 4.5:1 was also assumed, together with 50% decomposition of H_2O_2. Additionally, a reaction conversion of 10% over 24 hours at elevated temperature without a catalyst was assumed for a reaction identified in the calculation notes. These assumptions were used to estimate the consumption of reagents and the distribution of chemical species within the leaching system.

For the downstream separation and precipitation stages, reagent requirements were calculated using assumed reaction stoichiometry and conversion efficiencies. A 1:1 ratio was assumed when determining the oxalic acid requirement for Ni, Co, and Mn oxalate precipitation. The washing-water requirement for the recovered NMC oxalate solids was assumed to be 1 kg of water per kg of captured solids, while the wet filter cake was assumed to contain 30 wt% water before drying. For lithium carbonate recovery, sodium carbonate addition was calculated based on the stoichiometric requirement for lithium precipitation, and 1 kg of washing water per kg of lithium carbonate was assumed. Liquid retention within separated solids was also considered, although the exact retained quantities require further verification.

For the off-gas treatment and neutralization stages, complete neutralization was assumed for the applicable reaction. The off-gas treatment calculations were also informed by a referenced acid-gas removal method. Collectively, these assumptions established a consistent basis for the material balance calculations and the estimation of process stream flow rates. However, the assumed reaction conversions, separation efficiencies, and moisture contents should be validated against experimental results or published data during subsequent stages of process design.

Energy Balance

The energy balance was developed using the first law of thermodynamics. For each unit operation, the utility duty was determined from the sensible enthalpy change, heat of reaction, latent heat associated with evaporation, heat losses, and mechanical shaft work, as applicable. Equations 26 to 34 present the calculation methods, and the results are summarized in ***Table 21***. The terms included in each unit balance depend on the operations taking place; terms that do not apply are set to zero.

Overall Energy Balance:

$$\dot{Q}_{utility} = \Delta\dot{H}_{sensible} + \dot{Q}_{rxn} + \dot{Q}_{evap} + \dot{Q}_{loss} - \dot{W}_{p} \quad \textbf{[Equation 26]}$$

Mass Flow Rate:

$$\dot{m} = (m_{annual} \times 1000\ kg/tonne) / (8760\ h/yr \times 3600\ s/h) \quad \textbf{[Equation 27]}$$

Sensible Enthalpy Change:

$$\Delta\dot{H}_{sensible} = \Sigma\ \dot{m}_i\ C_{p,i}\ (T_{out,i} - T_{in,i}) \quad \textbf{[Equation 28]}$$

Chemical Reaction Heat Rate:

$$\dot{Q}_{rxn} = \Sigma\ \dot{\xi}r\ \Delta H_{rxn,r} \quad \textbf{[Equation 29]}$$

Scrubber and Evaporative Latent Heat:

$$\dot{Q}_{evap} = \dot{m}_{water,evap}\ \Delta H_{vap} \quad \textbf{[Equation 30]}$$

Solid Dryer Theoretical Heat Duty:

$$\dot{Q}_{theoretical} = \Delta\dot{H}_{solid} + \Delta\dot{H}_{liquid,water} + \dot{Q}_{latent,evap} + \Delta\dot{H}_{superheat,vapour} \quad \textbf{[Equation 31]}$$

Solid Dryer Utility Duty:

$$\dot{Q}_{utility,dryer} = \dot{Q}_{theoretical} / \eta_{thermal} \quad \textbf{[Equation 32]}$$

Centrifuge Acceleration Power:

$$\dot{W}_{acceleration} = \tfrac{1}{2}\, \dot{m}_{total}\, (\omega r)^2 \quad \textbf{[Equation 33]}$$

Estimated Centrifuge Motor Power:

$$P_{motor} = (\dot{W}_{acceleration} \times F_{friction}) / \eta_{motor} \quad \textbf{[Equation 34]}$$

Sign convention: $\dot{Q}_{utility}$ is positive when heat is supplied to the unit, $\dot{Q}_{loss}$ is positive for heat lost to the surroundings, and $\dot{W}_p$ is positive for shaft work delivered to the process. Reaction enthalpy is negative for exothermic reactions and positive for endothermic reactions. In **Equations 33 and 34**, ω is angular velocity in rad/s and r is the effective radius in metres; the resulting power is in watts when SI units are used.

Assumptions

The process is assumed to operate continuously at steady state for 8,760 h/yr. Stream flow rates represent baseline operating throughputs. Most units operate at approximately 1 atm; the NMC oxalate vacuum filter (VF-101A/B) operates at 0.3 bar absolute pressure, corresponding to an approximate pressure difference of 0.69 atm when atmospheric pressure is taken as 1.0 bar. Net shaft work is neglected for the chemical leaching reactor and gravity filtration in the simplified balance. Heats of dilution are neglected.

Because the process streams are predominantly aqueous, a representative liquid-water heat capacity of 4.184 kJ/(kg·°C) is used for aqueous streams. Other assumed heat capacities are 3.95 kJ/(kg·°C) for caustic scrubbing solution, 1.04 kJ/(kg·°C) for nitrogen purge gas, and 2.01

kJ/(kg·°C) for water vapour. Mass-weighted heat capacities for dry precipitate solids range from 0.50 kJ/(kg·°C) for CuS to 1.31 kJ/(kg·°C) for Li_2CO_3.

Decanter centrifuges (C-101A/B/C/D) are estimated using an empirical friction multiplier of 1.45 and motor efficiency of 88%. The vacuum filtration unit (VF-101A/B) uses a preliminary slurry-feed basis of 100 kg/h and assumes 20 wt% liquid retention in the wet cake. The filtration step is treated as adiabatic, with no chemical reaction or evaporation; the wet cake (Stream 28) and filtrate (Stream 31) are therefore assumed to leave at the same temperature. The 100 kg/h basis should be reconciled with the plant-wide mass balance before final equipment sizing.

Standard reaction enthalpies used in the preliminary balances are −176.8 to −271.4 MJ/kmol for reductive leaching, −124 MJ/kmol for CuS precipitation, −100 MJ/kmol for Fe/Al hydroxide precipitation, −30 MJ/kmol for NMC oxalate precipitation, and −103.5 MJ/kmol for caustic off-gas scrubbing. These are treated as exothermic reactions. Lithium carbonate precipitation is assigned +18.1 MJ/kmol and treated as endothermic. Each reaction enthalpy must be applied consistently with its corresponding reaction extent and stoichiometric basis.

Mechanical agitation is estimated at 10.0–15.0 kW per vessel, and ambient heat losses are estimated at 5.0–25.0 kW depending on the vessel and operating temperature. Centrifuge motor power is estimated at 1.84 kW at 2,838 rpm using **Equations 33 and 34**; auxiliary pumps are estimated at 3.50–5.00 kW. The assumed overall thermal efficiencies are 75% for the NMC oxalate dryer (D-101, 105°C) and 70% for the lithium carbonate dryer (D-102, 240°C). These values remain preliminary design assumptions.

Note: **Equation 33** estimates kinetic power associated with accelerating the flowing material. **Equation 34** is an empirical motor-power estimate, not a full manufacturer-rated centrifuge power

model. The specified heat capacities, reaction enthalpies, heat losses, and efficiencies should be checked against their source data before final design.

Table 21. Energy Balance Calculation Summary

Equipment Tag	**Operating Temperature (°C)**	**Pressure/ pH**	**Sensible Heat $\Delta\dot{H}_{sens}$ (kW)**	**Reaction Heat $\dot{Q}_{rxn}$ (kW)**	**Latent Evaporative Heat $\dot{Q}_{evap}$ (kW)**	**Net Duty (kW)**
R-101A/B/C/D	50.0	1 atm, pH<0	116.08	-1,538.83	0.00	-1,422.75
C-101A/B/C/D	50.0	1 atm, 2838 rpm	0.00	0.00	0.00	1.84
R-102A/B	20.0	1 atm, pH 1.8	-3,465.55	-31.19	0.00	-3,491.74
T-101/V-101	35.0	1 atm	5.44	-14.45	11.61	7.29
R-103A/B	50.0	1 atm, pH 4.5	3,473.51	-50.00	0.00	3,428.51
R-104A/B	50.0	1 atm, pH 2.0	19.90	-150.00	0.00	-125.10
VF-101A/B	50.0	0.3 bar	0.00	0.00	0.00	5.00
D-101	105.0	1 atm, 24 h	54.85	0.00	268.38	430.98
R-105A/B	90.0	1 atm, pH 10.5	4,750.09	271.50	0.00	5,031.59
F-103A/B	100.0	1 atm	0.00	0.00	0.00	3.50
D-102	240.0	1 atm, 2h	25.87	0.00	44.20	100.10

VII. Challenges

Initial Process Selection and Change in Approach

At the beginning of the degree project, our group planned to investigate bioleaching as a method of recovering valuable metals from spent lithium-ion battery black mass. The approach was appealing because it offered the possibility of reducing chemical consumption. As we reviewed the available research and began developing the process, however, we found that there was not enough information to support a reliable design at our proposed processing capacity. Most of the studies we encountered focused on laboratory experiments, and translating those results into an industrial process would have required several assumptions that we could not confidently justify.

One of the main difficulties was establishing the nutrient feed needed to maintain microorganisms. We also had trouble identifying all the compounds that could be produced during microbial activity and accounting for them in the process. Recovering and reusing microorganisms presented another concern. We considered membrane filtration as a possible recovery method, but it was unclear whether filtration alone would provide adequate separation and allow the microorganisms to be recycled effectively. In addition, fluorine-containing compounds in the black mass, particularly lithium fluoride (LiF), raised concerns about toxic conditions and the possible inhibition of microbial activity. These issues made the bioleaching route increasingly difficult to model and scale up.

After discussing the limitations of the original approach, we decided to move forward with chemical leaching using sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2). Chemical leaching is a more established method in hydrometallurgical battery recycling, and we were able to find more useful experimental results and operating conditions for this route. The change gave us a

stronger basis for developing the material balances, selecting equipment, and continuing with the process design. Although it did not eliminate the need for assumptions, it reduced some of the uncertainty associated with the biological system.

Black Mass Feed Composition

Another major challenge was determining a suitable composition for the black mass feed. We needed a complete composition to carry out material balances and represent the feed in Aspen Plus, but many published studies reported only the concentrations of the main recoverable metals. Information on oxygen, carbon, binders, electrolyte residues, and minor impurities was harder to obtain, particularly for Canadian black mass.

We selected an NMC111 black mass reference from a University of Toronto research group, based on material supplied by a Canadian recycler. The study reported measured concentrations for important elements, including nickel, cobalt, manganese, lithium, aluminum, copper, and iron. This provided a useful starting point because the material was relevant to the battery chemistry and Canadian context of our project. However, the published elemental measurements did not by themselves give us every chemical species needed for the process simulation.

To complete the feed basis, we had to estimate certain components, including oxygen, and make assumptions about how some elements were distributed among compounds. We also considered materials such as graphite, polyvinylidene fluoride (PVDF), lithium hexafluorophosphate ($LiPF_6$), and lithium fluoride (LiF). The composition was then adjusted to total 100 wt%. Some of these estimates were developed with AI assistance (Google Gemini) when suitable published values were unavailable. We recognize that these values are design assumptions rather than verified

measurements, and that they introduce uncertainty into reagent calculations, impurity tracking, and predicted recovery rates.

Research Limitations and Industrial Scale-Up

Finding research that could be applied directly to industrial design remained difficult throughout the project. Many papers provided useful recovery efficiencies or reaction conditions, but their experiments were performed with small quantities of material under controlled laboratory conditions. They often did not include enough information to determine full-scale equipment requirements, continuous operating conditions, recycle arrangements, or utility demands. We also found relatively little Canadian-specific information that matched our selected feedstock and proposed process.

To address this gap, our group decided to explore by contacting an industrial battery recycling company, formerly known as Li-Cycle, to introduce our project and ask whether they might be willing to share practical insight. Our goal is to better understand industrial operating conditions, process scalability, and potential areas for optimization. At this stage, this is a proposed outreach effort rather than a confirmed collaboration.

Process Modelling and Design Challenges

The change to chemical leaching made it easier to find supporting leaching data, but the overall process still required substantial modelling and design work. In Aspen Plus, the mixture of dissolved metals, aqueous electrolytes, solids, and impurities made component selection and thermodynamic modelling challenges. We also had to decide how best to represent precipitation, filtration, washing, gas treatment, and recycle streams using the available simulation blocks. Some

process steps could not be described directly by a single block and required simplifying assumptions or calculations outside the simulation.

These modelling difficulties were closely related to the gaps in literature. When information such as separation efficiency, product moisture content, reaction conversion, or equipment operating conditions was missing, we had to identify a reasonable design basis and document it clearly. Convergence problems and simulation errors also required us to revisit stream specifications and the way individual units were connected. As a result, the process flow diagram, stream tables, and simulation were developed through repeated revisions rather than in one continuous step.

Overall Reflection

The most important lesson from this stage of the project has been the difference between demonstrating that a recovery method works in a laboratory and designing a process that could operate at an industrial scale. Changing from bioleaching to chemical leaching was a practical decision based on the information available to us and the requirements of the project. Establishing a representative feed composition and making consistent assumptions have been just as important as selecting the recovery method itself. Going forward, we intend to improve the design by checking the assumptions against published data where possible, clearly identifying remaining uncertainties, and seeking industry input on scale-up and operating conditions.

VIII. Conclusions & Future Works

Acknowledgement

We would like to express our sincere gratitude to our degree project supervisors, Dr. Leila Pakzad and Dr. Sudip Rakshit, for their guidance, valuable feedback, and continued support throughout the development of this project. Their expertise and recommendations have been instrumental in shaping our understanding of hydrometallurgical processes and guiding the design of the proposed recovery system.

We would also like to acknowledge the Department of Chemical Engineering at Lakehead University for providing the academic foundation and resources necessary to undertake this project.

Finally, we extend our appreciation to all group members for their collaboration, dedication, and contributions toward the development of the proposed hydrometallurgical process for recovering valuable materials from spent lithium-ion battery black mass.

Nomenclature

Table 22. List of Abbreviations, Notations & Chemical Symbols, and Greek Letters & Mathematical Symbols and Their Definitions Used in the Hydrometallurgical Process Design Report.

Abbreviations	
LIB	Lithium-Ion Battery
EV	Electric Vehicle
NMC	Nickel-Manganese-Cobalt
NMC111	Nickel-Manganese-Cobalt Cathode Material (1:1:1 Molar Ratio)
PFD	Process Flow Diagram
PLS	Pregnant Leach Solution
DI	Deionized
CSTR	Continuous Stirred-Tank Reactor
RDVF	Rotary Drum Vacuum Filter
S/L	Solid-to-Liquid Ratio
MVR	Mechanical Vapor Recompression
Notations & Chemical Symbols	
Al	Aluminum
C	Carbon
Co	Cobalt
Cu	Copper
Fe	Iron
Li	Lithium
Mn	Manganese
Ni	Nickel
H_2SO_4	Sulfuric Acid
H_2O_2	Hydrogen Peroxide
H_2O	Water

Na_2S	Sodium Sulfide
NaOH	Sodium Hydroxide
Na_2CO_3	Sodium Carbonate
$H_2C_2O_4$	Oxalic Acid
CuS	Copper Sulfide
H_2S	Hydrogen Sulfide
N_2	Nitrogen
O_2	Oxygen
Li_2CO_3	Lithium Carbonate
NiC_2O_4	Nickel Oxalate
CoC_2O_4	Cobalt Oxalate
MnC_2O_4	Manganese Oxalate
$Ca(OH)_2$	Calcium Hydroxide
$CaSO_4 \cdot 2H_2O$	Calcium Sulfate Dihydrate (Gypsum)
Na_2SO_4	Sodium Sulfate
Greek Letters & Mathematical Symbols	
ΔP	Pressure Difference
η	Efficiency
ρ	Density
ΔH	Enthalpy Change

IX. References and Citations

[1] Lombardo, T., Analyst, E. T., Ivo Walinga, Officer, C. P., McKimm, R., Manager, C. P., Bennett, S., Timur Gül, & Officer, C. E. T. (2026, September 7). *The rise of lithium-ion batteries – analysis - IEA*. IEA. https://www.iea.org/commentaries/the-rise-of-lithium-ion-batteries

[2] Ricin, N. K., Valantine Takwa, L., Yoro, K. O., Makambo, J. B., Christopher, C. N., Ibegbulam, C., Eterigho-Ikelegbe, O., Kingsley, U., & Jen, T.-C. (2025). Lithium-ion batteries and the future of sustainable energy: A comprehensive review. *Renewable and Sustainable Energy Reviews*, *223*, 115971. https://doi.org/10.1016/j.rser.2025.115971

[3] Mulimbayan, F., et al. (2026). Green leaching and selective recovery of Ni, Co, and Mn from lithium-ion battery black mass with citric acid and H_2O_2. ACS Sustainable Chemistry & Engineering, 14(27), 11764–11772. https://doi.org/10.1021/acssuschemeng.6c00069

[4] Antuñano, N., et al. (2025). Direct production of cathode active material from black mass of spent lithium nickel manganese cobalt oxide batteries: Integrating and optimizing Li_2CO_3 recovery for enhanced economic and environmental performance. Resources, Conservation & Recycling Advances, 28, Article 200291. https://doi.org/10.1016/j.rcradv.2025.200291

[5] Recycling lithium-ion batteries: Technologies, environmental impacts, and hydrometallurgical membrane processes. (2024). Membranes, 14(12), Article 277. https://pmc.ncbi.nlm.nih.gov/articles/PMC9025773/

[6] Lithium recovery and conversion from wastewater produced by recycling of Li-ion batteries via two-stage electrodialysis. (2025).

https://www.researchgate.net/publication/390077355_Lithium_Recovery_and_Conversion_from_Wastewater_Produced_by_Recycling_of_Li-Ion_Batteries_via_Two-Stage_Electrodialysis

[7] Alfa Laval. (n.d.). Decanter centrifuge for solid-liquid separation. https://www.alfalaval.ca/products/separation/centrifugal-separators/decanters/

[8] Biswal, B. K., & Balasubramanian, R. B. (2023). Recovery of valuable metals from spent lithium-ion batteries using microbial agents: A review. Frontiers in Microbiology, 14, Article 1197081. https://doi.org/10.3389/fmicb.2023.1197081

[9] Roy, J. J., et al. (2023). Closed-loop graphite recycling from spent lithium-ion batteries through microbial leaching. ACS Sustainable Chemistry & Engineering, 11(17), 6567–6577. https://doi.org/10.1021/acssuschemeng.2c07262

[10] Kang, X., Cai, L., Li, Y., Gao, X., & Bai, G. (2022). Investigation on the separation performance and multiparameter optimization of decanter centrifuges. Processes, 10(7), Article 1284. https://doi.org/10.3390/pr10071284

[11] Pei, C., Ma, L., Xia, T., & Li, S. (2023). Research on the optimization and application of the washing dechlorination process for municipal solid waste incineration fly ash. ACS Omega, 8(4), 4081–4091. https://doi.org/10.1021/acsomega.2c07094

[12] Baust, H. K., Hammerich, S., König, H., Nirschl, H., & Gleiß, M. (2024). Resolved simulation of the clarification and dewatering in decanter centrifuges. Processes, 12(1), Article 9. https://doi.org/10.3390/pr12010009

[13] Estay, H., Barros, L., & Troncoso, E. (2021). Metal sulfide precipitation: Recent breakthroughs and future outlooks. *Minerals, 11*(12), 1385. https://doi.org/10.3390/min11121385

[14] Isaksson, A., Anaya Garzon, J., Strandkvist, I., & Sundqvist Öqvist, L. (2025). Nickel and cobalt recovery from spent lithium-ion batteries via electrodialysis metathesis. *Membranes, 15*(4), 97. https://doi.org/10.3390/membranes15040097

[15] Occupational Safety and Health Administration. (n.d.). *Hydrogen sulfide: Evaluating and controlling exposure.* https://www.osha.gov/hydrogen-sulfide/evaluating-controlling-exposure

[16] Barnes, C. M., & Hanna, T. (2016). *Method and apparatus for removing acid-gases from hydrocarbon-bearing saltwater solution* (US Patent Application No. US20160200592A1). https://patents.google.com/patent/US20160200592A1/en

[17] Remediation of sulfides in produced waters of the oil and gas industry using hydrogen peroxide. (2024). *Water, 16*(14), 1987. https://doi.org/10.3390/w16141987

[18] Desai, B. O., Day, D. R., & Ctvrtnicek, T. E. (1983). *Logan Wash field treatability studies of wastewaters from oil shale retorting processes.* Monsanto Research Corporation. EPA Contract 68-03-2801. https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=30000FKI.TXT

[19] Lee, E., Rout, P. R., Kyun, Y., & Bae, J. (2020). Process optimization and energy analysis of vacuum degasifier systems for the simultaneous removal of dissolved methane and hydrogen sulfide from anaerobically treated wastewater. *Water Research, 182*, 115965. https://doi.org/10.1016/j.watres.2020.115965

[20] Chernyaev, A., Wilson, B. P., & Lundström, M. (2021). Study on valuable metal incorporation in the Fe–Al precipitate during neutralization of LIB leach solution. *Scientific Reports, 11*, 23283. https://doi.org/10.1038/s41598-021-02019-2

[21] Chernyaev, A., Zhang, J., Seisko, S., Louhi-Kultanen, M., & Lundström, M. (2023). Fe^{3+} and Al^{3+} removal by phosphate and hydroxide precipitation from synthetic NMC Li-ion battery leach solution. *Scientific Reports, 13*, 21445. https://doi.org/10.1038/s41598-023-48247-6

[22] National Institute for Occupational Safety and Health. (n.d.). *NIOSH pocket guide to chemical hazards: Sodium hydroxide.* https://www.cdc.gov/niosh/npg/npgd0565.html

[23] National Institute for Occupational Safety and Health. (n.d.). *NIOSH pocket guide to chemical hazards: Hydrogen peroxide.* https://www.cdc.gov/niosh/npg/npgd0335.html

[24] Gao, R., et al. (2020). Recycling $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$ material from spent lithium-ion batteries by oxalate co-precipitation. Vacuum, 173, Article 109181. https://doi.org/10.1016/j.vacuum.2020.109181

[25] Schmitz, D., et al. (2024). Co-precipitation of metal oxalates from organic leach solution derived from spent lithium-ion batteries (LIBs). Metals, 14(1), Article 80. https://www.mdpi.com/2075-4701/14/1/80

[26] Angellinnov, F., et al. (2025). Optimization of NMC811 synthesis via oxalate coprecipitation method for lithium-ion battery cathode. Metalurgi, 39, 79–88. https://doi.org/10.55981/metalurgi.2024.759

[27] Yoon, H., et al. (2024). Carbon dioxide utilization in lithium carbonate precipitation: A short review. Environmental Engineering Research, 29(3), Article 230553. https://doi.org/10.4491/eer.2023.553

[28] Zheng, Q., et al. (2026). Efficient preparation of battery-grade Li_2CO_3 from spent battery leachate via integrated solvent extraction and direct precipitation. Green Chemistry, 28(21), 8988–9001. https://doi.org/10.1039/d6gc00941g

[29] Tarleton, E. S., & Wakeman, R. J. (2007). Solid/liquid separation equipment. Solid/Liquid Separation, 1–77.

[30] Sivakumar, T., Vijayaraghavan, G., & Vimal Kumar, A. (2011). Enhancing the performance of rotary vacuum drum filter. International Journal of Advanced Engineering Technology, 2(4), 41–47.

[31] Dittmer, D., Andary, M., Diaz, F., & Friedrich, B. (2025). Evaluation of filter cake washing processes in hydrometallurgical battery recycling of lithium-ion batteries to optimize recoveries. Metals, 15(11), 1262.

[32] Do, M. P., Roy, J. J., Cao, B., & Srinivasan, M. (2022). Green closed-loop cathode regeneration from spent NMC-based lithium-ion batteries through microbial leaching. ACS Sustainable Chemistry & Engineering, 10(8), 2634–2644.

[33] Johnston, T. G., & Kochhar, A. (2020). Process for recovering materials from rechargeable lithium-ion batteries (Australian Patent No. AU2020202401A1). Australian Patent Office.

[34] Mazur, A., Pushkaran, S. A., Wedin, S., Wreland, L. R., & Marques, P. F. (2026). Recovery of lithium and metal oxalates: A simultaneous precipitation-based approach for circular battery recycling. Separation and Purification Technology, 403, 138716.

[35] Lappi, J. (2023). Filtration and washing of precursor cathode active materials. LUT University.

[36] Gao, W., Li, C., Xu, C., & Wang, D. (2016). Experimental study on water separation process in a novel spray flash vacuum evaporator with heat-pipe. Desalination, 386, 39–47.